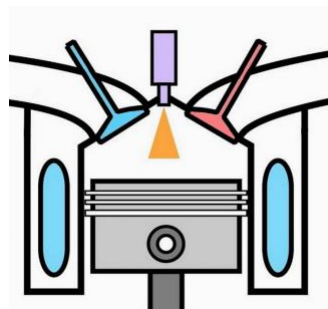
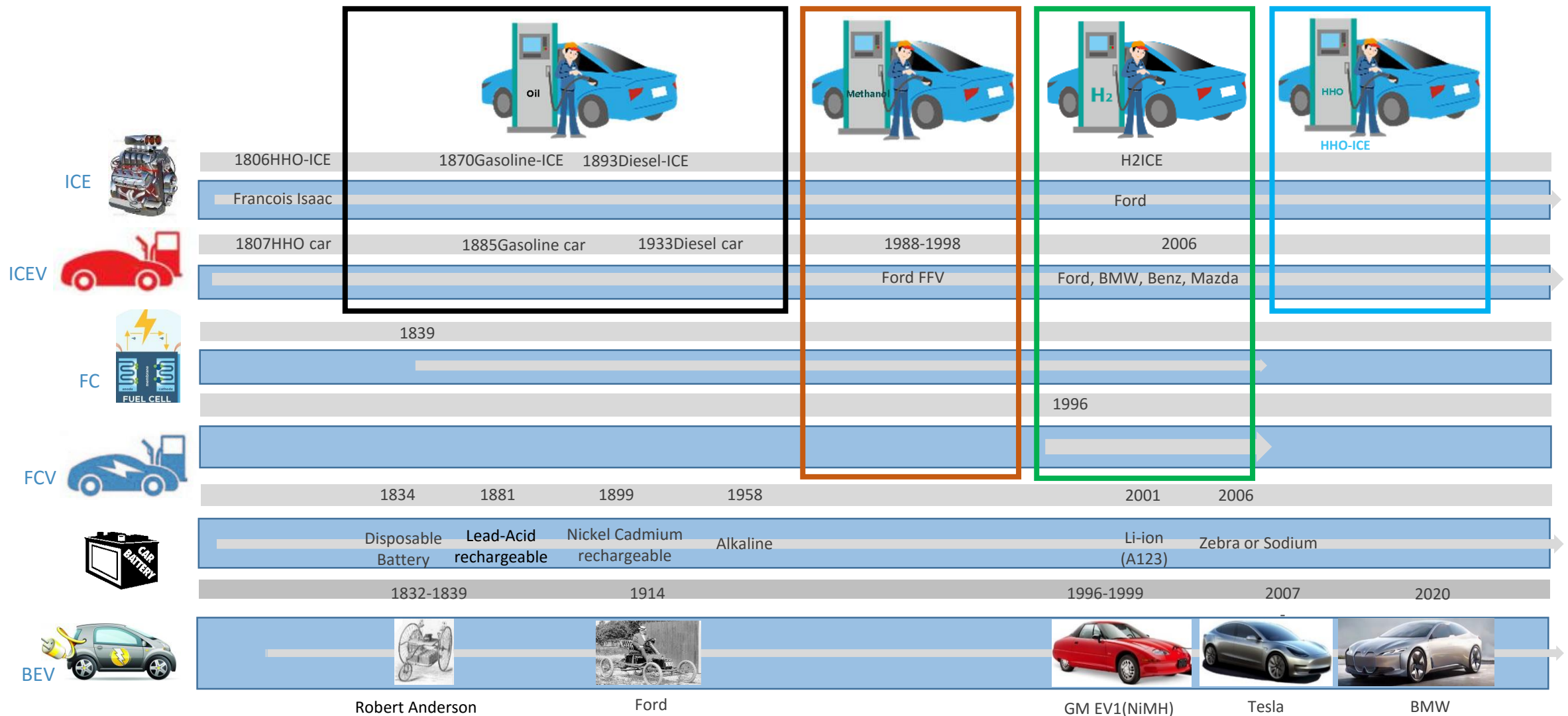




HHO1.0-现有技术和眼前商机

Smart Engine For Smart Vehicle To Smart World



The power system is roughly classified into three categories, ICEV, **BEV**, **FCV**, and the last two seems to be the future direction. My study shows the HHO-ICE will beat the BEV & FCV.

January 28th 2013, Ford, Daimler, and Renault-Nissan announced they would pool research into fuel cell technology. In their joint press release, they would produce fuel cell vehicle in 2017.

<https://www.cnet.com/roadshow/news/ford-mercedes-benz-and-nissan-target-2017-for-fuel-cell-vehicles/>

In 2017, former chief technology officer Raj Nair said fuel cell technology is developing in a very disjointed way.

<https://www.carscoops.com/2018/06/ford-daimler-planning-end-fuel-cell-joint-venture/>

June 14th 2018, Ford and Daimler planning to end their fuel cell joint venture. Ford will take fuel cell stack development in-house, as well as leverage the supply base, and close Automotive Fuel Cell Cooperation Corp. by summer 2018.

<https://ca.reuters.com/article/technologyNews/idCAKBN1J92YE-OCATC>

We are not clear the story behind it



Hydrogen – as the Fuel of the Future, which have advantages over battery-powered cars: they offer shorter refill times and longer driving ranges.

We acknowledges stating that the vehicles could be “an important long-term solution for improving energy security and diversifying our energy sources.” We has been working on the technology since 1997.

We acknowledged there are hurdles to overcome: **the price of fuel cell car** and a **lack of infrastructure** to support the cars.

This presentation will show a chance to overcome these two hurdles.

In order to fix the H₂ Fuel Cell Cost issue, Ford solution is to develop the H₂-ICE:

1997, Ford Motor Company Scientific Research Laboratory started the project to design and develop a practical, low-cost hydrogen fueled internal combustion engine (H₂ICE) vehicle.

2001, Ford introduced P2000 H₂ICE uses a modified version of the Zetec 2.0-liter gasoline engine. The engine's efficiency is improved by 25 to 30 percent over its gasoline counterpart.

2006, Although Ford's work on the new HySeries fuel-cell hybrid continues to gather momentum, teams within the company have made great strides to bring hydrogen fuels to internal combustion engines—specifically, the Tri-Flex fueling 6.8-liter Triton V10.

To run solely on hydrogen, engineers reinforced pistons, connecting rods and valve seats, but the biggest change is the addition of a supercharger. Running at an 18- to 20-psi boost, the supercharger helps the motor put out 235 hp at 4000 rpm and 310 lb-ft at 3000 rpm. These figs. are meager compared with a naturally aspirated gas V10 (305 hp at 4250 rpm, 410 lb-ft at 3250 rpm).



Supercharger only engages when the Tri-Flex engine is consuming hydrogen fuel

Unique hydrogen fuel system includes separate fuel rails and injectors

Same things happened in Europe.

2007, BMW produced about 100 H2 ICE vehicles named BMW Hydrogen 7, it has been modified to also allow for the combustion of hydrogen as well as gasoline.

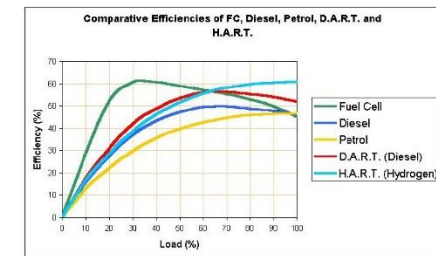
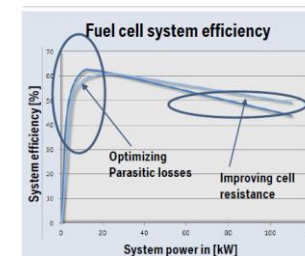
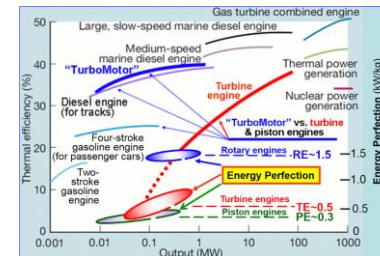
Some agency figure out that the efficiency of the H2 ICE too low.

1. The efficiency the car had on hydrogen is about 5.6 mpg (50 l/100km) on average, mostly due to the difference in energy density between petrol and hydrogen. So the H2 Fuel Cell Vehicle have the same issue.
2. The H2-ICE system efficiency is about the 40~47%, which is lower than the Fuel Cell of 60%(peak).

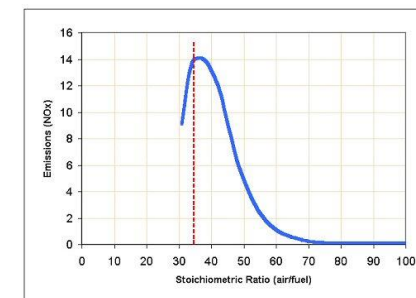


Fuel Consumption Table for the BMW Hydrogen 7

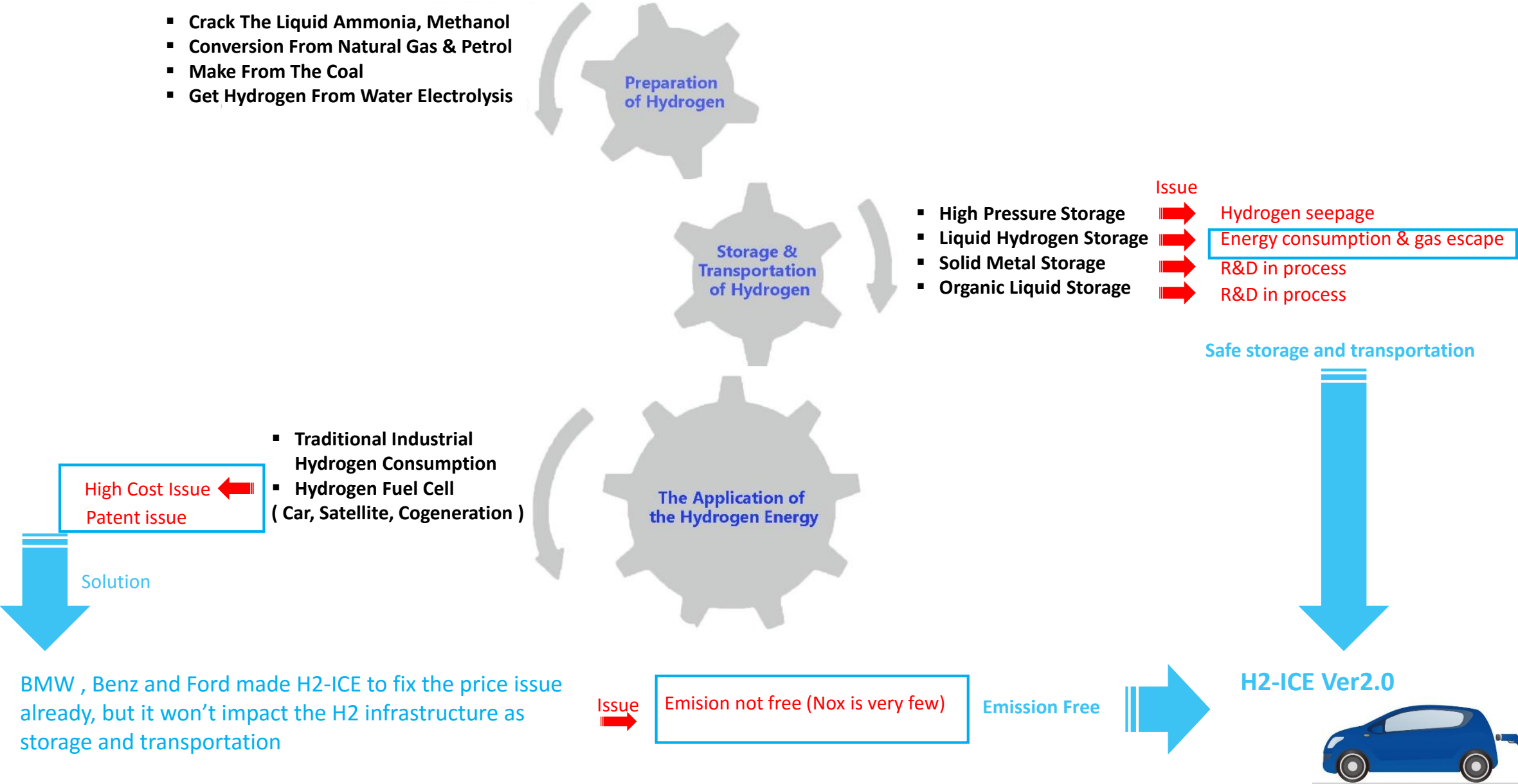
Gasoline (petrol)			Hydrogen		
L/100 km	Imp. mpg	US mpg	L/100 km	Imp. mpg	US mpg
13.9	20.3	16.9	50.0	5.6	4.7



3. the V12 hydrogen engine achieves nevertheless a power output of 191.2 kW (256-hp) but same petrol engine can meet 438-hp ($\text{H}_2 + \text{O}_2 + \text{N}_2 \rightarrow \text{H}_2\text{O} + \text{Nox}$, in order to reduce the Nox, typically hydrogen engine are desgined to use about thice as much air as theoretically required for comple cobustion. At this air/fuel ratio, the formation of Nox is reduce to near zero. Unfortunately, this also reduces the output to about half that of a similarly sized gasoline engine. That's why Ford also use the supercharger but is still less power output.



Hydrogen Energy Industry Chain

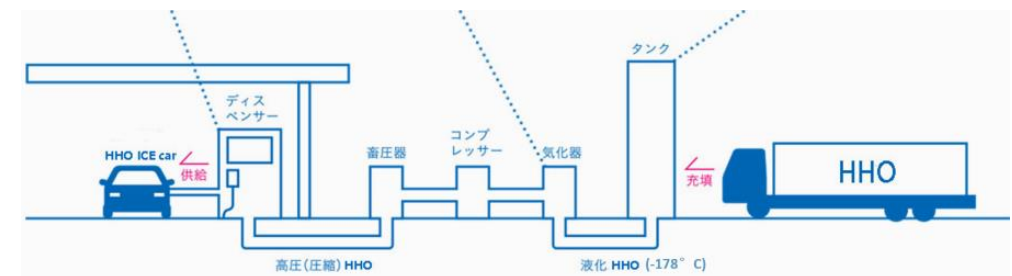
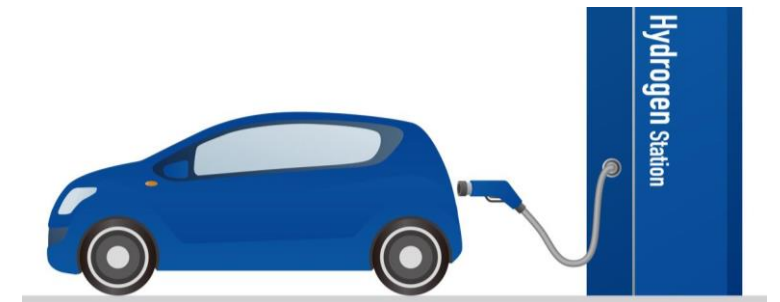


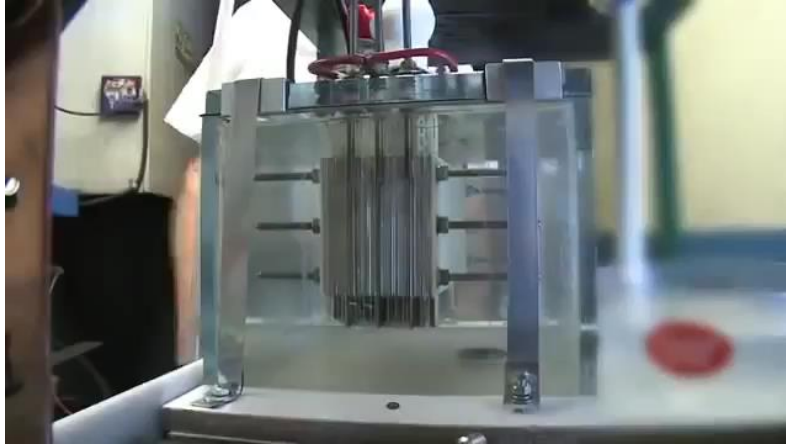
HHO VS H₂

For the ICE, The fuel determines the characteristics of the ICE, search for the ideal fuel becomes the ultimate goal of the development of the ICE.

Hydrogen is considered to be is the ideal fuel to the ICE.

My study shows the ideal fuel to the ICE should be the HHO.





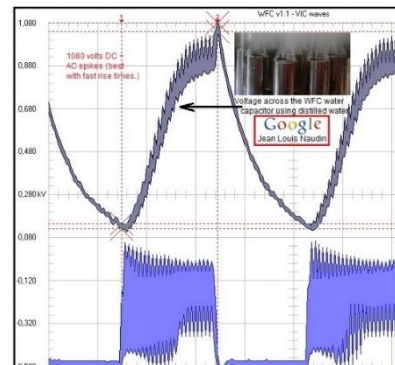
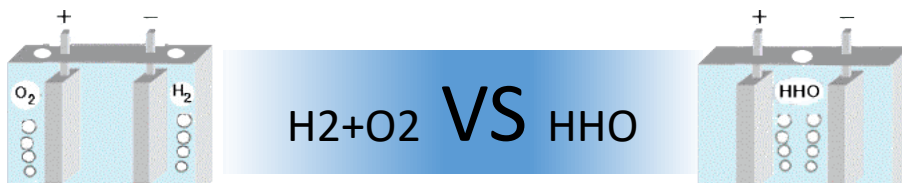
Video-How to get the HHO



Video-interesting character of HHO

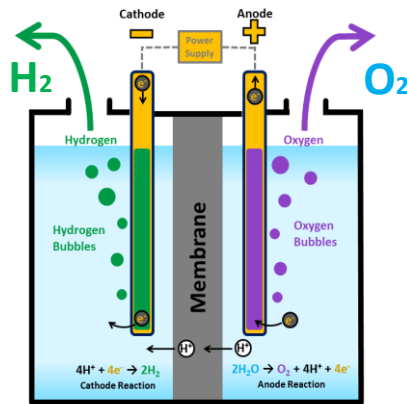
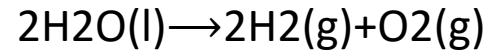
Unlike the conventional electrolysis, there are two necessary condition to get the HHO producing is:

1. the We need to keep the electrodes very close. Smaller than 1mm is better.
2. We need to use 'pules electrolysis' equipment.

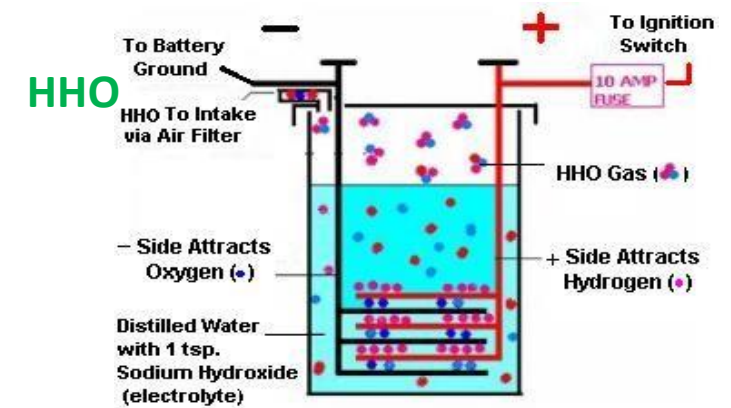
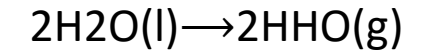


What is HHO?





The water electrolysis to H₂ and O₂



The water electrolysis to HHO

H₂+O₂ VS HHO

Definition 1 (https://en.wikipedia.org/wiki/Oxyhydrogen#HHO_gas)

Wikipedia explain that the HHO (oxyhydrogen or the Brown's gas) is a mixture with 2:1 molar ration of H₂ and O₂ gases, the same proportion as in the water. It is a mixture gases (2H₂ + 1O₂).

Definition 2 (<http://tophogascar.com/what-is-hho-gas/>)

HHO gas is a newer form of gas that is created by using 'pules electrolysis' equipment. Being made up of magneclules that contain hydrogen gas, HHO is connected to lone atoms that are oxygen atoms, by "magneclular bonds".



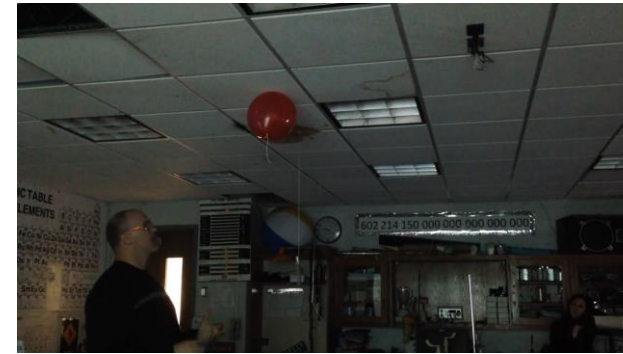
Which one is correct?



Video-LPG Explosion



Video-H₂ Explosion



Video-2H₂+O₂ Explosion



Video-HHO Explosion

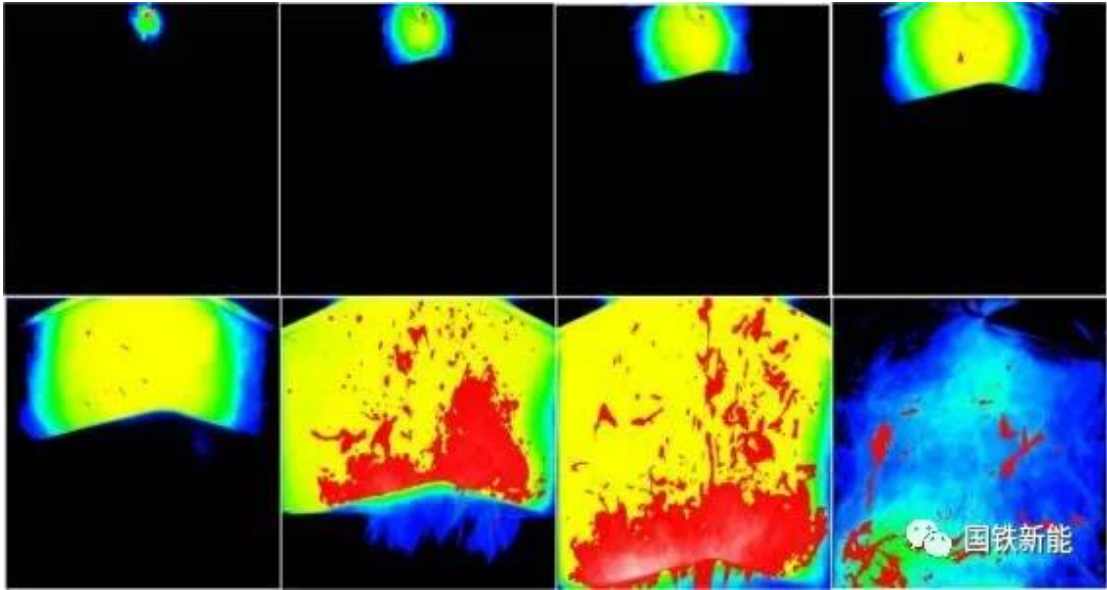
The video shows that **HHO explosion speed higher than the mixture gas of oxy-hydrogen (2H₂+O₂)**, and the mixture gas (2H₂+O₂) higher than the hydrogen in the atmosphere, and hydrogen higher than the LPG (petrol assume).

Gasoline+Air	X	<i>The Explosion Speed</i>
H ₂ +Air	3X	
2H ₂ +O ₂	12X	
HHO	18X	

This is an analogy, not an exact description,

What is HHO?

The HHO explosion speed is significantly higher than the mixture gas of 2:1 ratio of H₂ & O₂, **HHO should not be the mixture of 2H₂ + O₂.**



在几秒钟内将沸点超过摄氏5500度的钨金汽化，真可谓“无坚不摧”

The hydrogen combustion temperature is related to the oxygen concentration and the hydrogen flow rate.

The maximum flame speed is the equivalent combustion in air: 2.65m/s

The equivalent flame temperature in air: 1430 °C

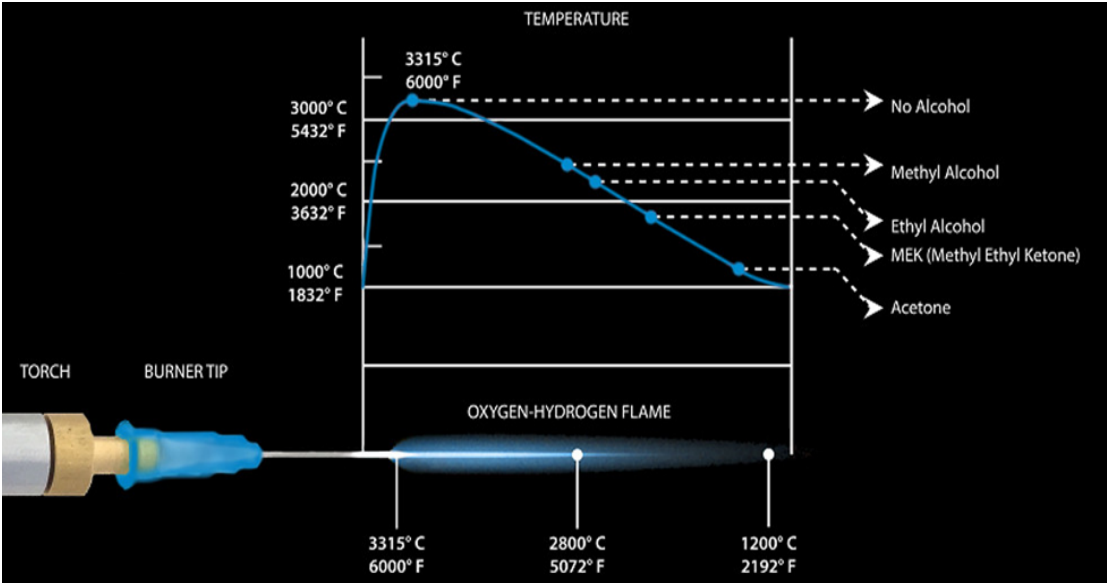
The maximum flame speed in the equivalent combustion in oxygen: 14.36m/s

The equivalent flame temperature in oxygen: 2830°C

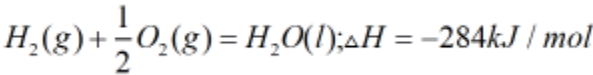
Acetylene

Hydrogen

物质	火焰温度	物质	火焰温度
乙炔焰（氧炔焰）	3000°C	丙烷火焰	1995°C
蜡烛	800~1400°C	金属镁	1900~2300°C
氢气	1150~1650°C	天然气（甲烷）	900~1100°C
木炭	750~1100°C		



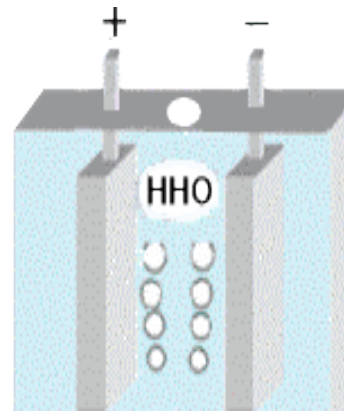
What is HHO?



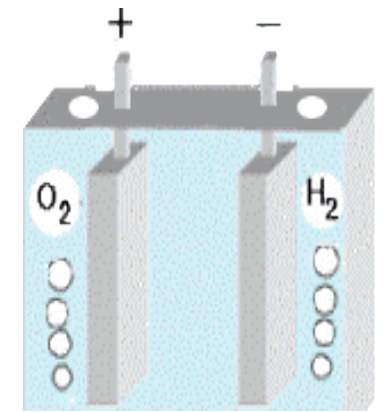
The temperature of the hydrogen flame in the air or in the oxygen is much lower than the HHO flame temperature. **HHO gas is not a mixture of H2 and O2 gases.**

On June 30, 2003, **Adsorption Research Laboratory of Dublin, Ohi**, measured the specific weight of the HHO gas and released a signed statement on the resulting value of **12.3 g/mol**. The same laboratory repeated the measurement on a different sample of the gas and confirmed the result. The released value of 12.3 g/mol is anomalous. In fact, the conventional separation of water into H₂ and O₂ produces a mixture of 2/3 H₂ and 1/3 O₂, which value should be **11.3g/mol**.

$$2/3 \text{ H}_2 + 1/3 \text{ O}_2 = (2 + 2 + 32)/3 = 11.3\text{g/mol.}$$

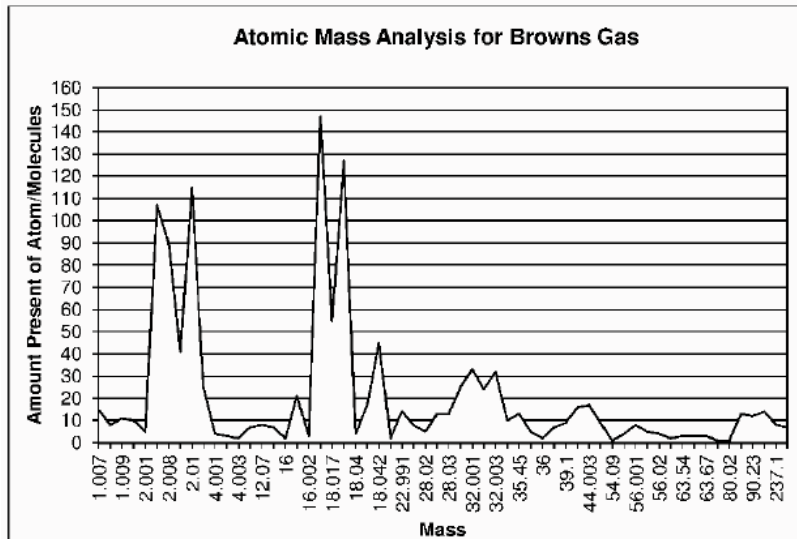


12.3 g/mol



11.3g/mol

HHO VS H₂ + O₂



What is HHO?

The Molecular weight of the HHO gas is different from the mixture of 2/3 H₂ and 1/3 O₂. It shows that the **HHO gas is not a mixture of 2H₂ + O₂**.



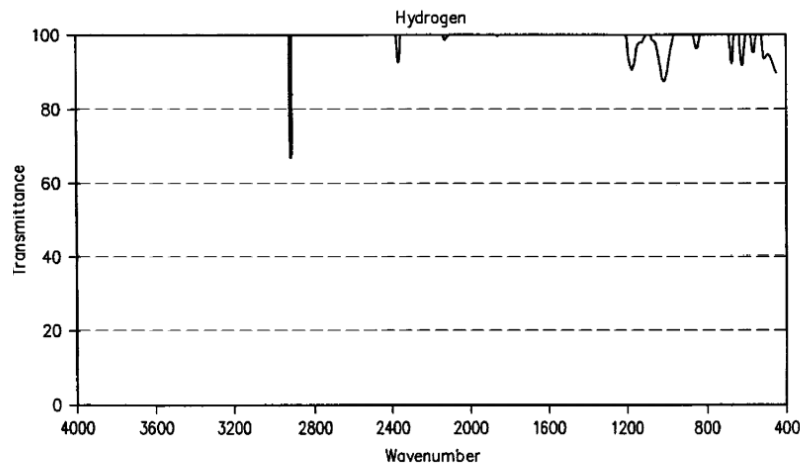
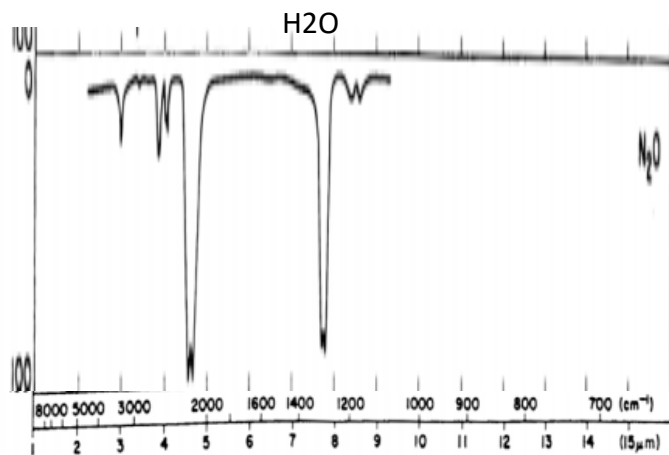


Fig. 2. The IR signature of a conventional H₂ gas made by the PdMA laboratory.

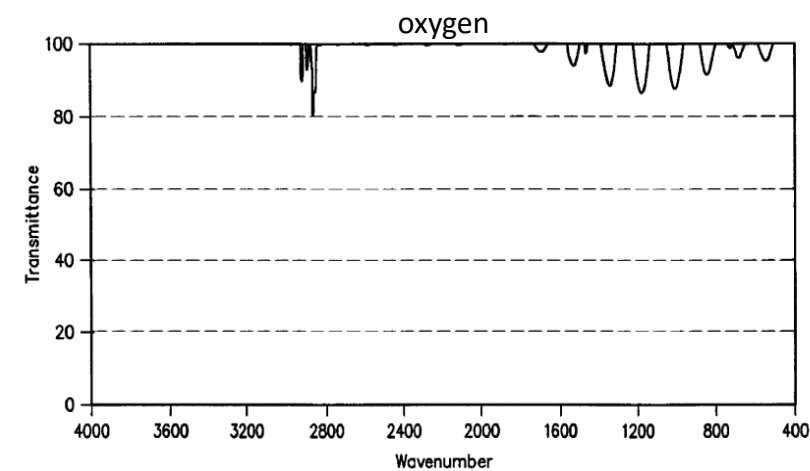
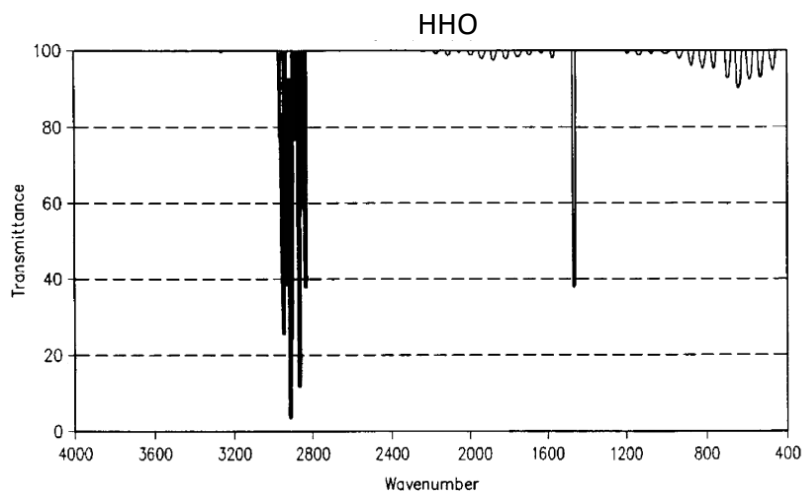


Fig. 3. The IR signature of a conventional O₂ gas made by the PdMA laboratory.



What is HHO?

The IR signature of the HHO gas made by the PdMA laboratory in Tampa, Florida shows it differ from the IR scans of H₂ and O₂, this scan shows that the **HHO gas is not a mixture of H₂ and O₂ gases.**



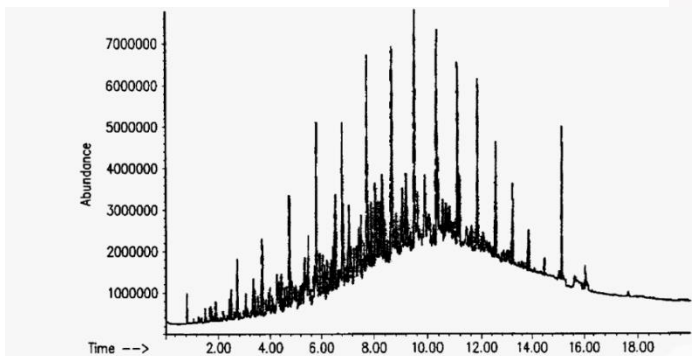


Fig. 5. A TIC of the GC-MS scans of conventionally sold diesel fuel made by Southwest Research Institute.

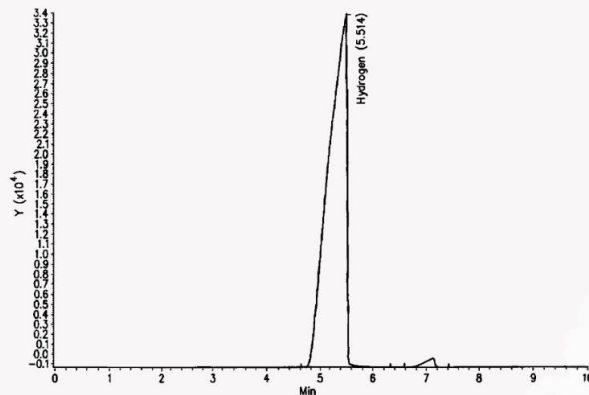


Fig. 7. A TIC of the GC-MS scans on the HHO gas made by Toxic LTD Laboratories showing the H₂ content of the HHO gas.

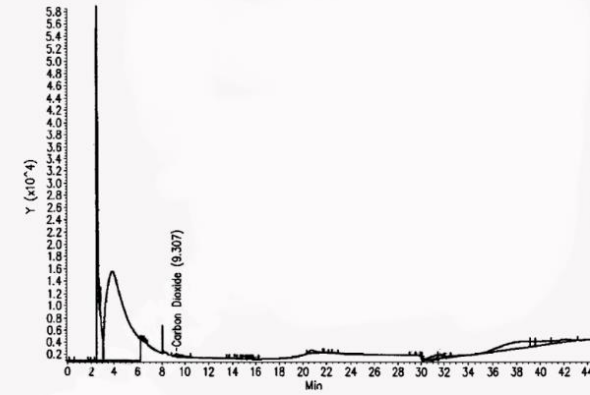


Fig. 8. A TIC of the GC-MS scans on the HHO gas made by Toxic LTD Laboratories showing the peaks belonging to H₂ and O₂, plus anomalous peaks.

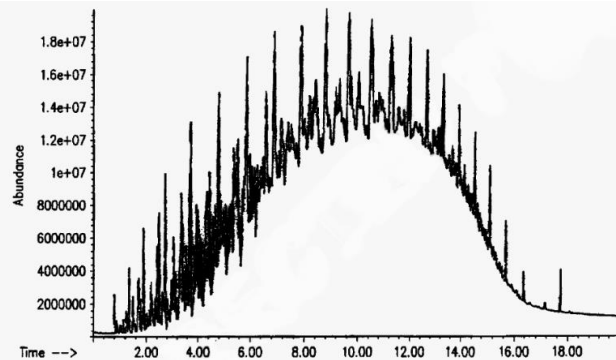
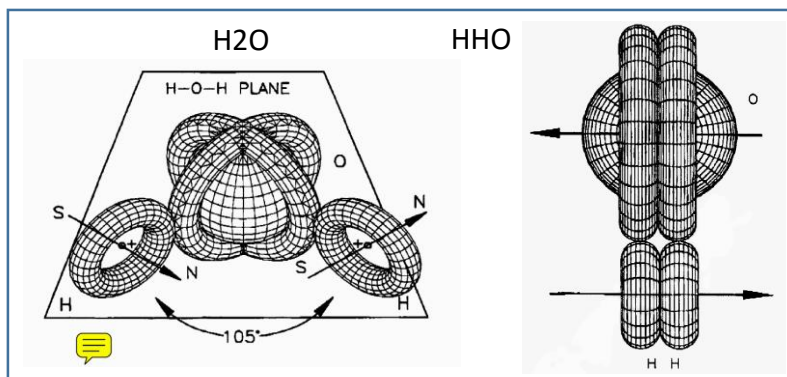


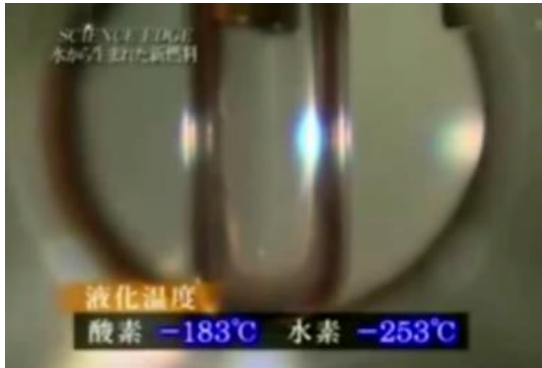
Fig. 6. A TIC of the GC-MS scans made by Southwest Research Institute on the same diesel fuel of Fig. 5 in which the HHO gas had been bubbled through, showing the alteration of the TIC both in shape as well as increased mass, thus indicating a new bond between diesel and HHO that cannot be of valence type (since HHO is gaseous and diesel is liquid. In any case, all valence electrons in both the gas and the liquid are used by conventional molecular bonds).



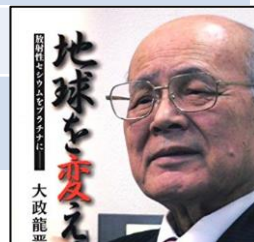
What is HHO?

The Southwest Research Institute have a research report about this unknown gas and infer the molecular structure of HHO should be the left fig. shows. **HHO gas is not a mixture of H₂ and O₂ gases.**





	Liquefied temperature	Energy consuming
H2	-253℃	40 MJ/kg (11.1KWh)
N2	-196℃	877KJ/Kg (0.24KWh)
O2	-183℃	3.6-4.5MJ/Kg (1KWh)
HHO/MHHO?	-178℃	2MJ/Kg (0.2~1KWh) ?
LNG	-162℃	510KJ/Kg (0.14KWh)

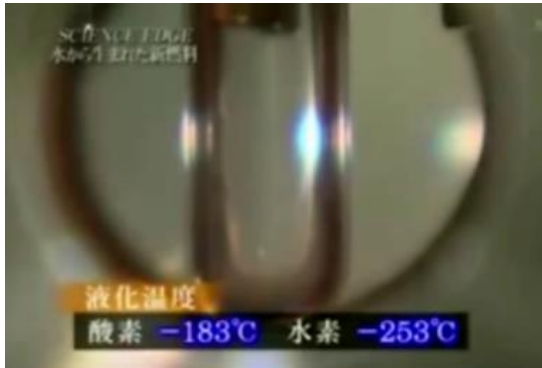


TOKYO
METROPOLITAN
UNIVERSITY

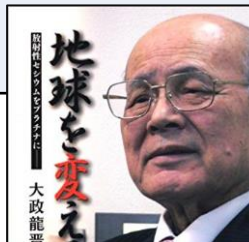
What is HHO?

The Liquefied temperature and the color of the MHHO gas researched by the TOKYO METROPOLITAN UNIVERSITY shows it differ from the liquefied temperature of H2 and O2, **HHO gas is not a mixture of H2 and O2 gases.**





	Oxygen Liquid	Hydrogen Liquid	Hydrogen Gas		MHHO Liquid
	L-O2- -196℃	L-H2- -253℃	35MPa	70MPa	L-MHHO- -178℃
Hydrogen Density	1410kg/m ³	70.8kg/m ³	15kg/m ³	30kg/m ³	167kg/m ³ (converted to Hydrogen content) ?
					1503kg/m³ (Liquid MHHO Density) ?
	2/3 LH2+1/3 LO2 = 517.2kg/m³				



The liquid MHHO contains 2.5 times hydrogen compare to the liquid hydrogen in volume.



TOKYO
METROPOLITAN
UNIVERSITY

What is HHO?



The liquid MHHO density researched by the TOKYO METROPOLITAN UNIVERSITY shows it differ from the liquid density of 2/3H₂ and 1/3O₂, **HHO gas is not a mixture of H₂ and O₂ gases.**

Conclusion 1

HHO is a poorly defined term, it is not exactly ($2\text{H}_2 + \text{O}_2$)



HHO gas is already in widely use



HHO gas is ideal fuel to ICE



What is HHO?

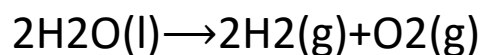
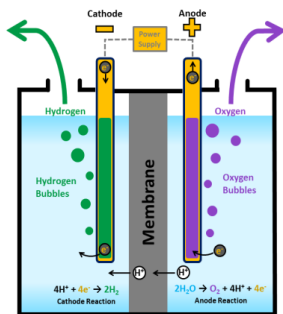


Conclusion 1

HHO is a poorly defined term, it is not exactly (2H₂+O₂)



权威派



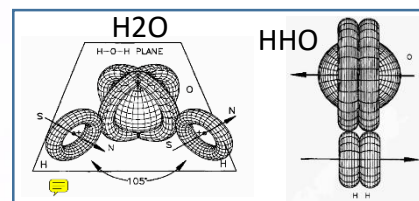
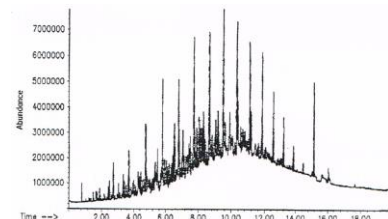
学术界



研发界



产业界



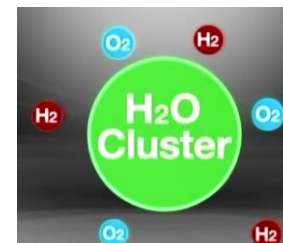
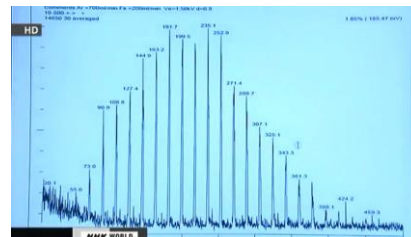
What is HHO?

吃不准



吃瓜群众

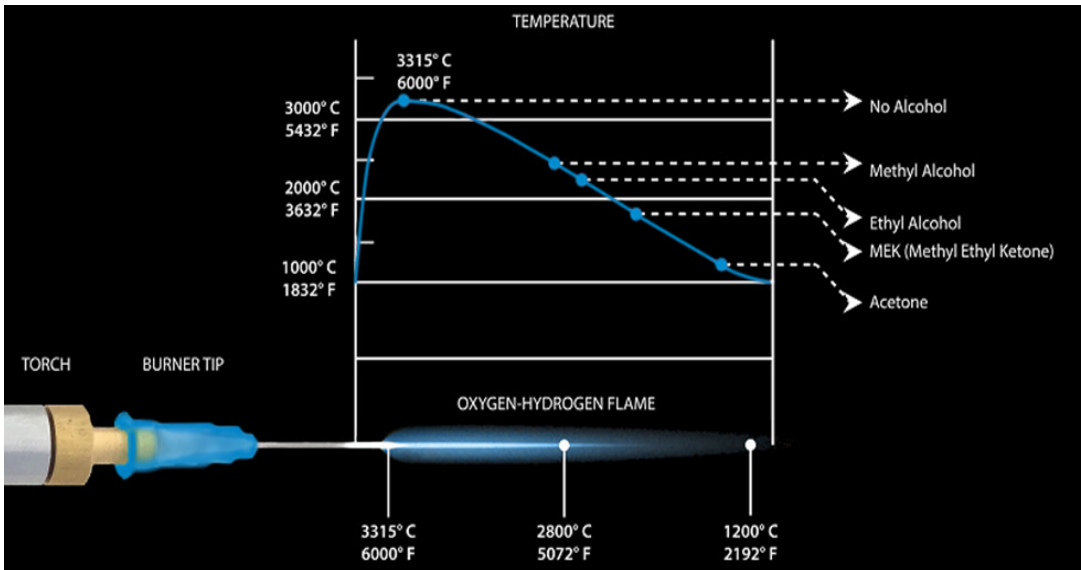
玄真派



HHO Is Widely Use In Metal Cutting

The HHO flame temperature is extremely high with low heat radiation, the combustion is pollution free, smoke free, which is much better than other fuel gas, that why the HHO gas is widely used in Metal Cutting.

Hydrogen flame temperature in air is only 2100 degree Celsius
Hydrogen flame temperature in Oxygen is only 2800 degree Celsius

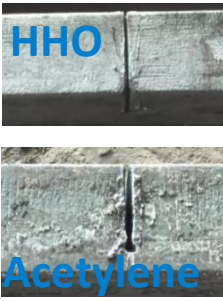
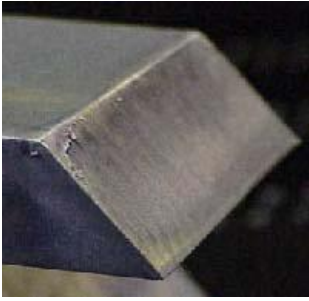


在几秒钟内将熔点超过摄氏6000度的钨金烧化，
切割别的材料真是削铁如泥（这个不是文学比喻）。

HHO Flame Underwater



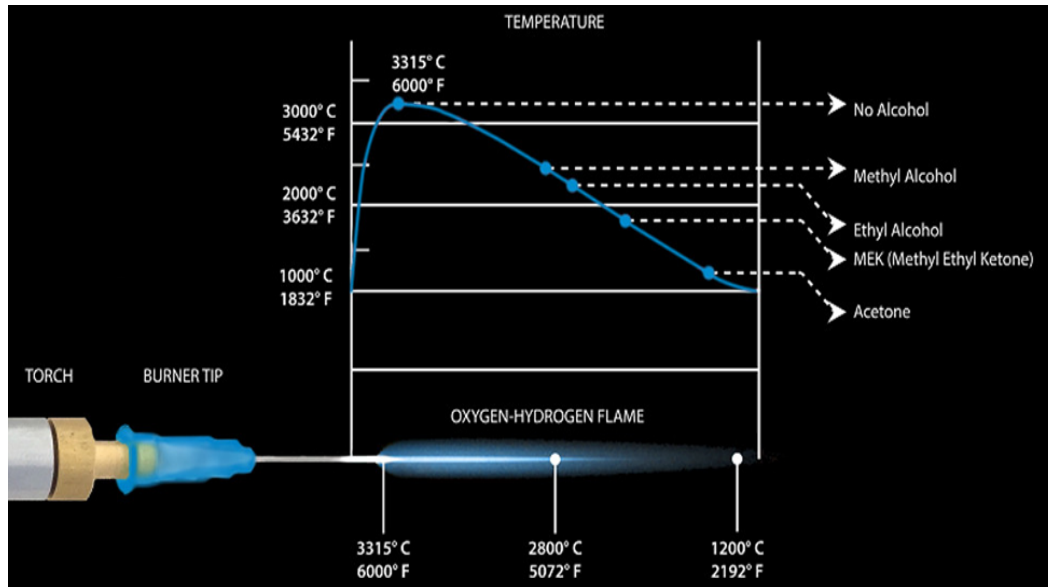
HHO Flame Underwater Cutting



Video-Metal Cutting

Video-Metal Cutting

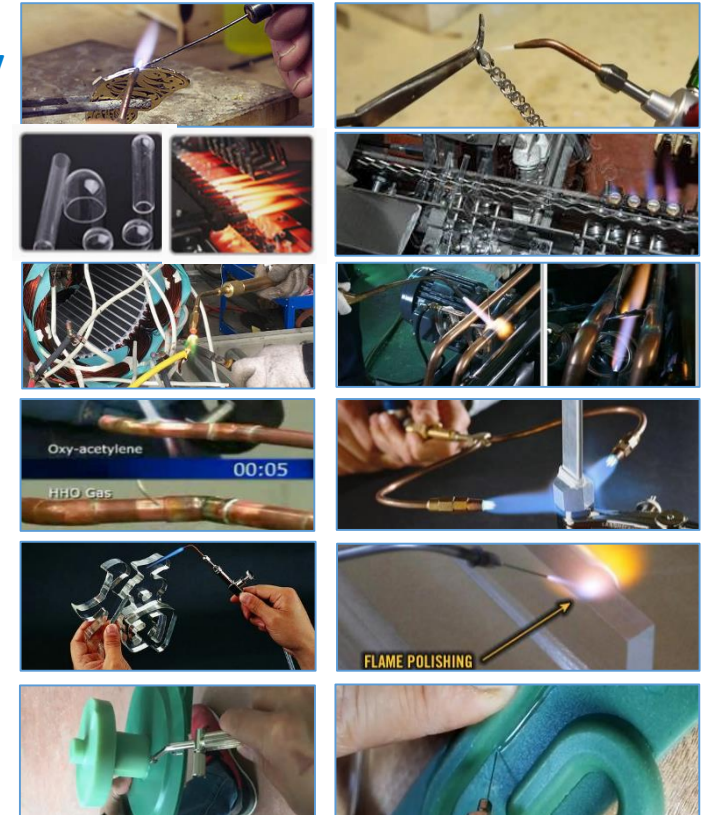
HHO Is Widely Use In Soldering



Zirconia hot pad is being welded to the steam pipe by HHO flame. This is an amazing property of the HHO's flame to weld firmly different materials together, like steel or glass to ceramic, etc., which no other gas or arc welder can do. (the hydrogen gas can not do this)

The HHO flame temperature is extremely high with low heat radiation, the combustion is pollution free, smoke free, which is much better than other fuel gas, that why the HHO gas is widely used in soldering place.

- Jewelry Manufacturing Industry
- Medical Device Industry
- EM Manufacturing Industry
- Non-ferrous metal Welding
- Advertising Industry
- Wax pattern soldering



EPOCH 友荃科技實業股份有限公司

The garbage collection rate is 100 percent. All the garbage is collected within a day.

1500°C=2732°F

Before the Oxy-hydrogen being added

the Oxy-hydrogen being added

All of the small size incinerators which were not able to burn at high temperature were shut down as they generated dioxins under 800°C (二恶英) .

HHO Is Widely Use by Boilers and Electric Generator

As A Combustion Improver

Mobile Incineration station



		
Fuel	Heavy oil + HHO gas	Heavy oil
Width	250-300mm	300-350mm
Length	1600mm	1200mm
Features	No soot	Generated soot
	Fire is Strong	Fire is spread and weak
	Bright white fire color	Blackish red fire color

HHO Boiler



HHO Stove

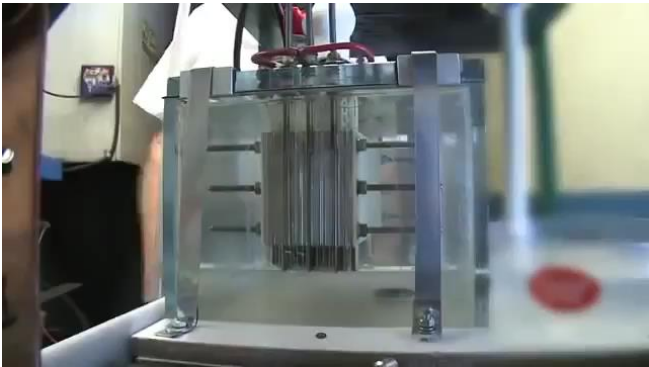


Video-HHO Infrared tube

With high flame temperature and faster flame propagation speed, it can be the ideal fuel for the mobile incineration station, and improve the burning in the Boiler and in the electric generator.

HHO Is Widely Use As Fuel Saver

HHO As A Fuel Saver (auxiliary combustion)



Video-Independent Agency Report 40% fuel efficiency improvement



Taiwan Customer use HHO Kit



American customer use HHO Kit



Japanese customer use HHO kit



Dr Soib (seated) explaining the HHO Generator. With him is USM vice-chancellor Prof Datuk Dr Omar Osman.

Cummins KTA-38
Eurofins test results onboard Hydrogen on demand generation.



Measurement	Without HHO		With HHO		Delta	Avoided emissions at 2500 running h
	% Vol	g/h	% Vol	g/h		
O ₂ (% Vol)	14.7	~	16.9	~	+14.97	~
CO ₂ (% Vol)	4.6	~	2.3	~	- 50.00	159 ton/year
CO (mg/ Nm ³)	679	784	504	453	- 25.77	828
NOx (mg/ Nm ³)	1645	1898	1332	1189	- 19.03	1773
PM (mg/ Nm ³)	187	215.8	41.2	37.1	- 77.97	446.75



Video-EPOCH HHO KIT

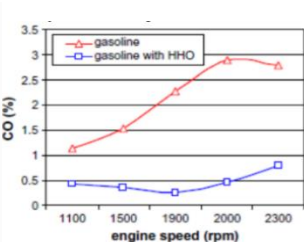
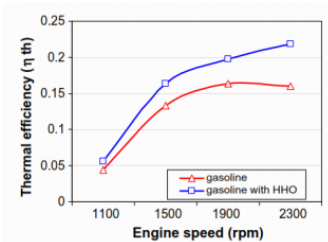


Figure 2: Engine speed vs CO

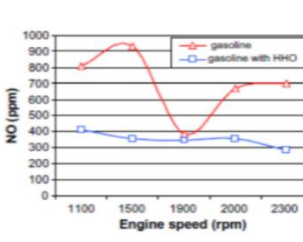


Figure 3: NO Vs Engine Speed

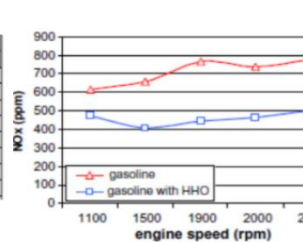


Figure 4: NOx Vs Engine speed

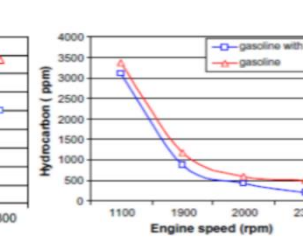


Figure 5: HC Vs Engine Speed

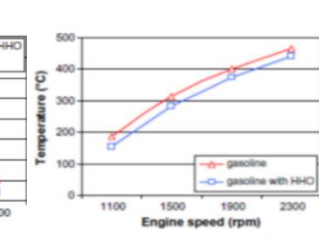


Figure 6: Temp Vs Engine Speed

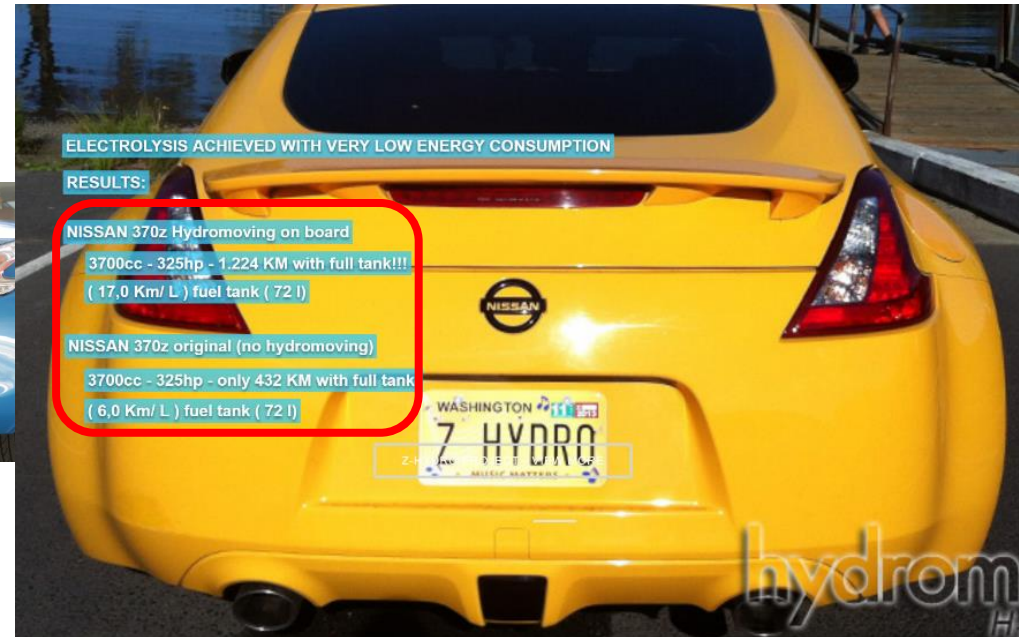
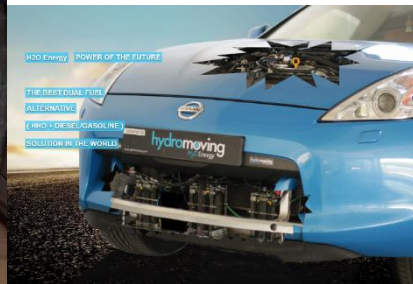
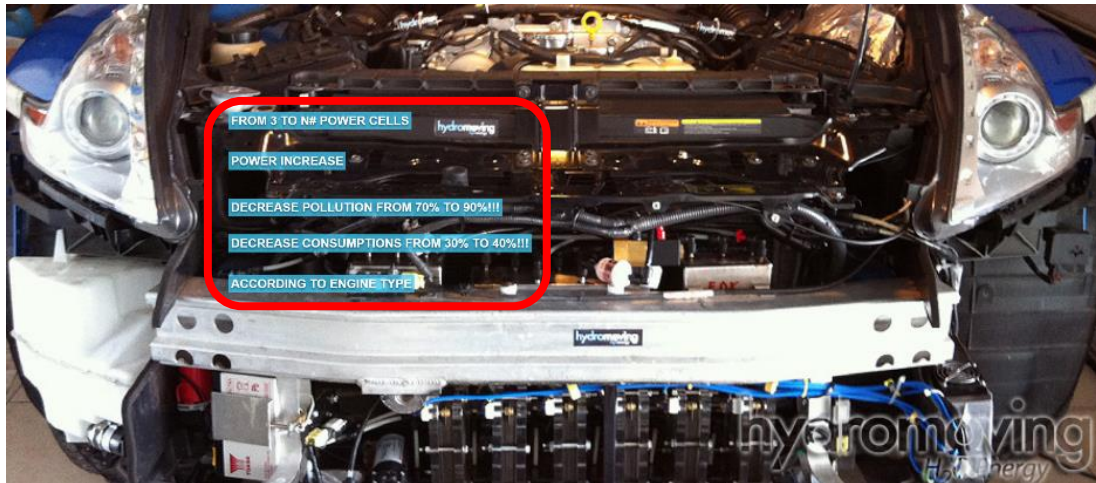
HHO act as a combustion improver to make the original fuel combustion more completely and cleanly.

“十二五（2011-2015）”期第一批全国重点推广公路水路交通运输节能产品（技术）目录

序号	产品名称 (型号 / 商标)	应用范围	生产或经销 单位名称	法人代表	联系人 及电话传真	地址及邮编	备注 (试验检测结果)
6	导流式汽车节能排气尾管 (STL050/圣天利)	汽油车	山东圣天利节能器有限公司	张 岩	李学军 Tel: 15949924676 Fax: 0534-5232996	山东宁津经济开发区 邮编: 253400	整车道路对比检测: 市区运行模式节油率 3.0%
7	生物复合纳米自组装燃油添加剂 (一鑫宝牌)	车用 柴油机	山东鑫海汽车配件有限公司	杜文军	张克民 Tel: 0635-3982268 Fax: 0635-3982269	山东省高唐县经济开发区鼓楼西路 邮编: 252800	发动机台架对比检测: 快速运行模式节油率 1.7%
8	HHO 车载氢氧机 (MCW-2000/ MCW)	车用 柴油机	东莞市鑫亚塑胶制品有限公司	张晓青	陈秋亭 Tel: 13829130090 Fax: 0769-85331810	广东省东莞市长安镇上角红棉路 6 号 邮编: 523878	发动机台架对比检测: 市区运行模式节油率 1.6%
9	原子自组装纳米球固体润滑剂 (WZ-QYCL-F-0.8-4.0L/—)	车用 汽油机	山东万众科技有限公司	宫志利	宫志利 Tel: 0533-6989601 Fax: 0533-6989621	山东省高青县经济开发区 邮编: 256300	整车道路对比检测: 市区运行模式节油率 3.1% 城间运行模式节油率 3.1% 快速运行模式节油率 3.0%
10	MD 环保节油剂 (MD-D/MD 图形)	车用、船用 柴油机	海南明德能源投资有限公司	徐德锋	康 明 Tel: 13647549818 Fax: 0898-68583617	海口市世贸东路 2 号世贸中心 F 栋 903 室 邮编: 570100	发动机台架对比检测 (车用): 快速运行模式节油率 1.7% 发动机台架对比检测 (船用): 推进特性平均节油率 1.5%
11	车用电子燃油优化处理器 (SPF-D/军丰)	车用 柴油机	广州市军丰节能科技有限公司	王伟林	王伟林 Tel: 020-87248950 Fax: 020-87248950	广东省广州市白云区京溪中路 17 号南 33 号 邮编: 510510	发动机台架对比检测: 市区运行模式节油率 2.0%



HHO Is Widely Use As Fuel Saver



NISSAN 370Z HYDROMOVING SYSTEM

Hydromoving compact System 6 cells on 3700cc engine.



BMW HYDROMOVING SYSTEM

HydroC4, four cells Compact model mounted on a 3 liters turbo diesel engine



SMART

Hydro3C, compact model installed on a Smart 1liter gas engine



NISSAN Z

HydroP6, Professional prototype model 6 cells installed on a 3.7 liters Nissan engine in USA.



NISSAN NV 200

Hydromoving compact system on 2000cc engine



Evoluzione GT

New installation and partner - Evoluzione GT



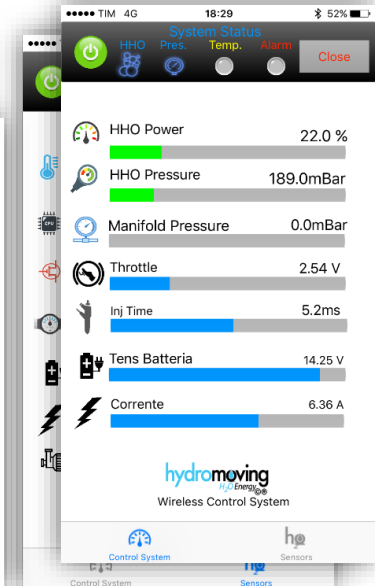
Mercedes ML

Hydromoving Compact System on Diesel 3.200cc Engine

The company HydroMoving Srl was founded through the efforts of the engineer Lorenzo Errico who, with his passion for cars, electronics and innovation with regard to "green technologies", involved enthusiasts in his own field giving life to the project HydroMoving.

Why Choose Us

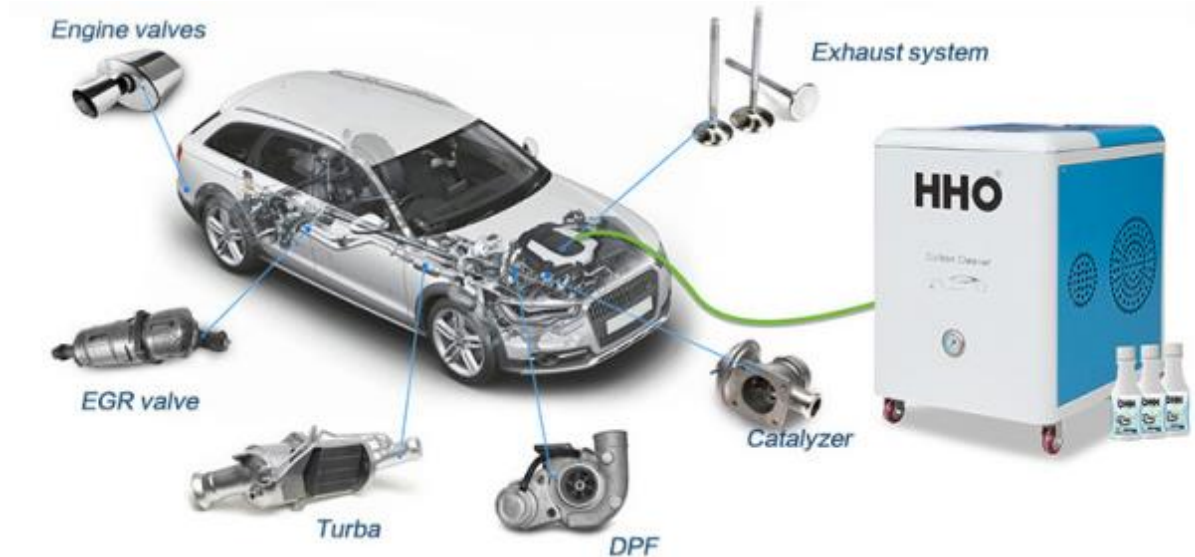
We are the only ones in the world to have a patented technology, tested and certified for the production of oxyhydrogen gas (gas produced from distilled water) on board and on-demand, with electronic injection in the combustion chamber of any type of ICE (gasoline / Diesel etc ..) that reduces polluting emissions by 90%. The system guarantees a reduction of CO2 one the GHG gas > = 30%.



<https://www.hydromoving.com>

HHO Is Widely Use in Engine Carbon Cleaning

HHO is widely use as Carbon Cleaner (auxiliary combustion)



Video-Engine Carbon Clean



HHO act as a combustion improver to make the original fuel combustion more completely and cleanly.

- 1) 1989年，澳大利亚科学家Yull Brown教授应中国北方工业公司(NORINCO)邀请来中国兵器工业第52研究所考察和技术交流；
- 2) 1990年2月，中国兵器工业第52研究所布朗气项目组成立；
- 3) 1991年5月，BN系列布朗气样机研制成功,并于第二年获国家级新产品奖；
- 4) 1994年12月，布朗气发生器通过内蒙古自治区科学技术成果鉴定；
- 5) 1995年，H1200布朗气发生器获国家级新产品奖、中国专利新技术产品博览会金奖；
- 6) 1996年，韩国B.E.S.T COREA CO.,LTD 公司金相南主席来公司考察；
- 7) 1998年12月，布朗气发生器通过GB/T19001-1994标准质量体系认证；
- 8) 1999年11月，H250B白金熔焊机获国家重点新产品奖（发证机关：中华人民共和国科学技术部、国家税务总局、中华人民共和国对外贸易经济合作部、国家质量技术监督局、国际环境保护总局）；
- 9) 2000年，布朗气项目被列为国家火炬计划项目；同年10月H3000型布朗气发生器正式上市销售；
- 10) 2002年，布朗气（氢氧气）产品被国家经贸委和国家税务总局列为第二批环保项目目录，享受相关优惠政策；我单位被科技局批准为《国家科技成果重点推广计划》项目“氢氧气发生器”的技术依托单位。
- 11) 2003年3月，布朗气项目进入中国兵器科学研究院宁波分院；
- 12) 2003年5月，推出采用布朗气技术的新型环保医用焚烧炉；
- 13) 2004年3月，新一代T80型火焰抛光机上市销售；
- 14) 2005年2月，宁波和利氢能科技有限公司成立，布朗气项目由和利公司进行完全市场化运作；
- 15) 2005年3月，公司与南京净之杰固体废物处理有限公司签订合作协议，将布朗气催化燃烧技术应用于QNK-15型控气式热裂解焚烧炉中进行生产性试验，获得突破性成果。
- 16) 2005年10月，采用专利技术的布朗气复合燃料燃烧器上市；
- 17) 2005年10月，氢氧气燃烧器荣获中国宁波科技创业大赛新秀奖；
- 18) 2006年5月，开展布朗气技术在大功率柴油发动机上的应用研究工作；
- 19) 2006年8月，布朗气燃油燃烧器系统通过宁波市科技局技术成果鉴定；
- 20) 2006年9月，公司总经理作为特邀嘉宾参加在北京人民大会堂举行的“第五届中国科学家论坛”；同年11月受邀参加第七届全国氢能学术会议，发表了学术论文并做了专题报告；
- 21) 2006年12月，新一代采用全封闭一体化电解槽结构的布朗气发生器正式上市；
- 22) 2007年3月，T1000型布朗气发生器批量投产；同年5月，T5000型布朗气发生器试制成功；
- 23) 2007年5月，公司作为特约嘉宾受邀参加我国第一届氢能展；
- 24) 2007年6月，T600型布朗气发生器批量投产；
- 25) 2007年9月，与扬子江药业签订合同，布朗气发生器首次应用在制药行业；
- 26) 2007年10月，公司参加上海国际广告机械展览，获得同行业高度好评；
- 27) 2007年11月，参加全国药品机械展，受到水针剂制药企业的普遍关注；
- 28) 2008年1月，与宝山钢铁集团公司达成连铸坯切割合作意向；
- 29) 2008年7月，在连铸坯切割领域与上海振华焊割公司签署长期战略合作协议；
- 30) 2008年9月，世界最大布朗气发生器T20000型试制成功，并通过宝山钢铁集团验收；
- 31) 2008年10月，通过了在大功率柴油发动机上的应用试验，并得到专家认可；
- 32) 2008年11月，与天津铜业合作，布朗气发生器T12500正式应用于冶金熔炼行业；
- 33) 2009年1月，T1500、T3000型布朗气发生器批量投产；
- 34) 2009年5月，产品出口英国，与英国DELICATE.ECO公司建立贸易合作关系，打开了欧洲市场；
- 35) 2009年10月，在汽车除碳技术上取得新突破，为产品拓展了一个全新领域；
- 36) 2009年11月，受国家氢能协会、标准化委员会委托，起草“氢氧发生器”国家标准；
- 37) 2010年9月，T系列产品通过欧盟CE认证，获得了欧盟国家的安检通行证；
- 38) 2011年1月，我公司先后主持起草《氢氧发生器安全技术规范》等4项国家标准的制定工作；
- 39) 2012年1月，产品通过交通部公路科学研究院检测认证，T1500系列产品开始应用于汽车领域；
- 40) 2012年4月，成为中国医药设备工程协会会员；
- 41) 2012年8月，我公司总工程师（研究员）成为全国氢能标准化技术委员会委员
- 42) 2013.5 入选中国汽车工业协会及中国汽保协会会员
- 43) 2014.2 布朗气分子除碳机获得北京汽保国际展览会特别推荐产品奖
- 44) 2015.1 三元催化养护机获得中国汽保协会技术贡献奖
- 45) 2015.2 通过ISO9001-2008质量体系认证
- 46) 2015.11 在中国新三板挂牌成功，股票代码：834311
- 47) 2016.10 获得宁波国家高新技术企业资质；
- 48) 2017.1 主持制定的国家标准《氢氧发生器安全技术要求》GB/T34539-2017正式发布实施；
- 49) 2018.6 喷油嘴清洗机HD10 获得年度全国“20佳维修工具”优秀奖。

HHO gas is widely use example



T2K氢氧焰焊机

型号T2K (RMB15000)

简介：
T2K氢氧焰焊机，可以焊接漆包线（引出线40平方以下）、铜管，2针手工石英玻璃焊接等等



浙江和利氢能科技股份有限公司
地址：浙江省宁波市国家高新区凌云路199号 邮编:315103
电 话：(86) 574—87902271/87902261 18157475745 传 真：
(86) 574—87902271
<http://www.browngas.cn>

Conclusion 2

HHO is a poorly defined term, it is not exactly ($2\text{H}_2 + \text{O}_2$) 🙄

HHO gas is already in widely use



HHO gas is ideal fuel to ICE

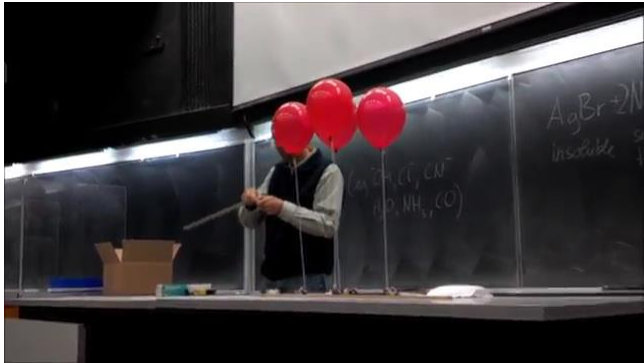


What is HHO?

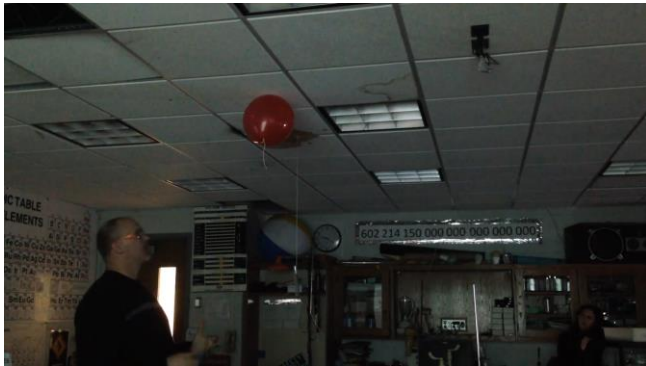


HHO Is Ideal Fuel to ICE

HHO VS H₂



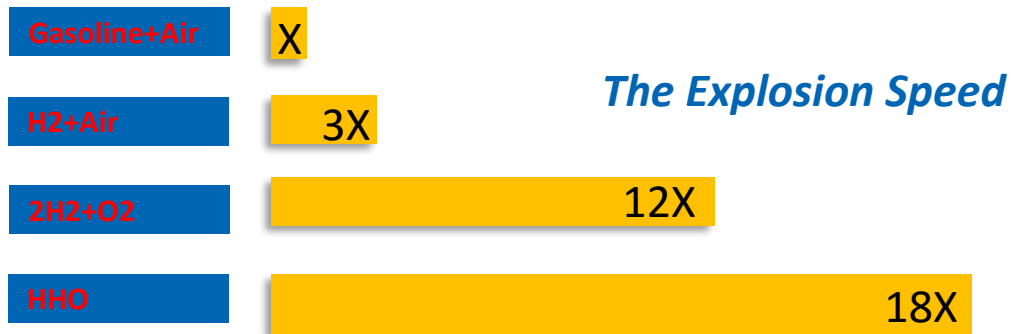
Video-Horse Power improvement (HHO VS H₂+Air)



Video-Horse Power improvement(HHO VS 2H₂+O₂)



Video-Explosion Speed (HHO VS H₂)



This is an analogy, not an exact description,

The information I collect shows that HHO flame propagation speed 3 times faster than H₂, 10 times faster than gasoline. And because the ratio of H:O is perfect 2:1 in atom level, the combustion is much better than the compressed air combustion with H₂ (turbine or supercharger). As the 78% of the air is N₂, the high burning speed will benefit to the accurate control of ignition timing.

HHO Is Ideal Fuel to ICE

HHO VS H_2

The HHO combustion is a kind of internal closed combustion with low temperature, It doesn't need the extra air, which is much different to the Hydrogen fuel. And the heat radiation is much low, the damage to the engine components should be much less as other fuel does(**need verification**)

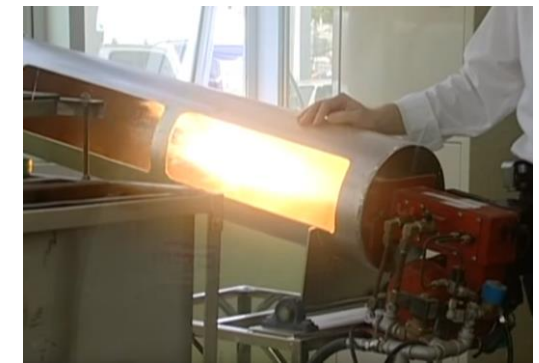


Video-HHO Explosion In Hand



Measurements of
perature and elect

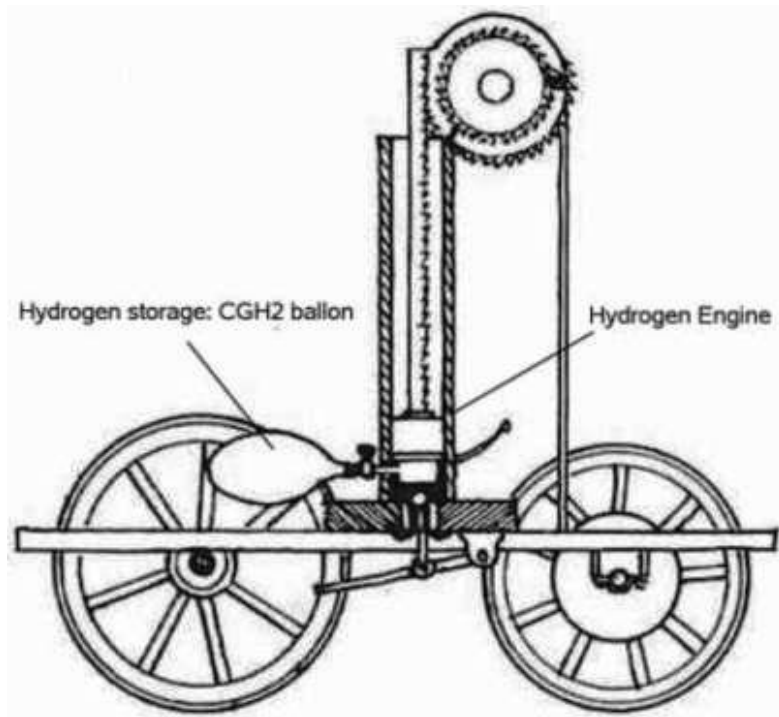
Temperatures		
Tungsten		
Melt	6192 F	3422 C
Vaporize	10031 F	5555 C
Browns Gas		
	266 F	130 C
Torches		
Acetylene	5972 F	3300 C
Hydrogen arc	7232 F	4000 C
Cyanogen	8477 F	4525 C
Dicyanacetylene	9009 F	4987 C



Video-HHO flame Temperature

Low heat radiation

HHO Is Ideal Fuel to ICE



1806 François Isaac de Rivaz (from Switzerland) makes the first internal combustion engine. The engine runs on mixture of hydrogen and oxygen (HHO)



Video-HHO MOTOR IN MUSEUM DE RIVAZ ENGINE

This is the actually first hho motor ever built and of course it is just sitting in a museum being wasted away. The motor has existed for decades and can produce electricity from small amounts off hho gas that drives the cylinder off the ground with small implosion and gravity pulls the piston back down for each cycle .The hho motor belongs to the people of the world not this museum. This motor uses very small needle amounts of hho gas to power this huge motor. The engine produces no carbon emission and no smoke runs on hho and water vapor is the only byproduct of operation.

HHO Is Ideal Fuel to ICE



Video-HHO MOTOR NVH improvement

Proof that HHO is a great oxidizer for most any fuel, plus it burn the fuel much more efficiently and cleaner.



Video-HHO engines don't need to exhaust their spent gases

HHO Is Ideal Fuel to ICE

The HHO can be used as a pure fuel to drive the engine, the exhaust gas temperature is much lower than the traditional one. Logically the engine can be simplified very much as the **air intake system can be removed** and the **cooling requirement can be reduced**. (need verification)

HHO VS Gasoline



Video-HHO/MHHO Run ICE



Video-HHO Run ICE



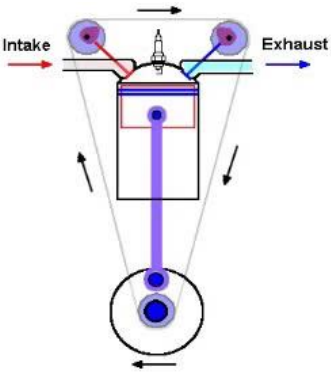
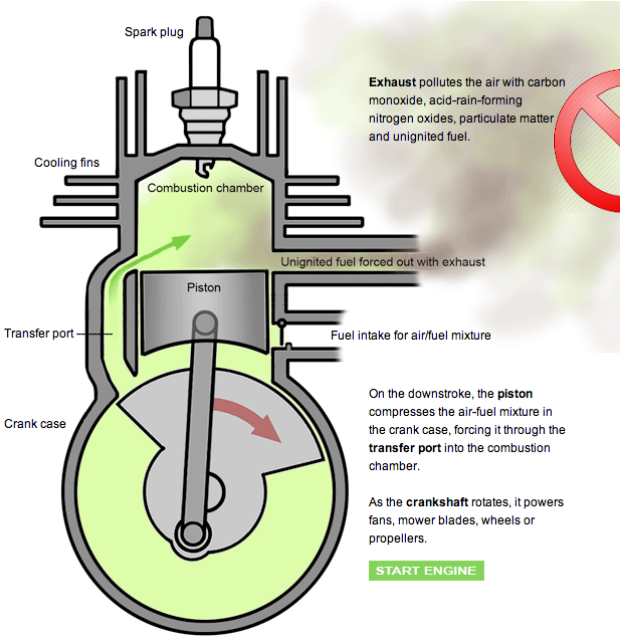
ICE Body Temperature @5min running
114 Fahrenheit = 45 Celsius

HHO Is Ideal Fuel to ICE

HHO VS H₂ VS Gasoline

More evidence shows that the HHO could be the ideal fuel for 2-strokes engine, which will get more output in lower engine speed and meet the emission free at same time. Or get the same output in small volume engine, which the Hydrogen fuel can not do it. (need verification)

	Gasoline	Hydrogen	HHO
4 strokes -ICE	YYY	YYYY	YYYYY
2 strokes-ICE	YY	YYYY	YYYYY
Stirling Engine	YY	YYYY	YYYYY
Turbojet engine	YY	YYYY	YYYYY



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Performance Characteristics of Oxy Hydrogen Gas on Two Stroke Petrol Engine

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²Assistant Professor, Dept of ME, Bannam Venkata Chaitanyam College of Engineering & Technology, Chittoor, AP, India

Abstract: In order to conserve petroleum fuels for future and to eliminate the above limitations, there is a need of alternative and innovative fuel. Amongst many alternative fuels available, oxy hydrogen gas and producer gas are under study in this research work. Electrolysis of water can give us hydrogen in form of Brown's gas (HHO) gas or oxy-hydrogen gas, which can be used as an alternative fuel for any internal combustion engine. Various methods for the generation of oxy-hydrogen gas are discussed in the coming paragraphs. An agricultural waste like rice husk, wood dust, and dried coconut leaves etc can also be used as an alternative fuel. An attempt has been made in this work to use alternative fuel in two stroke petrol engine. Our first aim was to select this project to use nonconventional fuel against conventional fuel which is becoming scarce and costly now days. The combustion of a hydrocarbon fuel with air produces mainly carbon dioxide (CO₂) and Water (H₂O). However, internal combustion engines are not perfectly efficient, so some of the fuel is not burned, which results in the presence of hydrocarbons (HC) other organic compounds, carbon monoxide (CO) and forming mainly nitric oxide (NO).
Keywords: Two Stroke Petrol engine, Electrolysis, Oxygen Enriched hydrogen(HHO) gas, emission characteristics.

1. Introduction

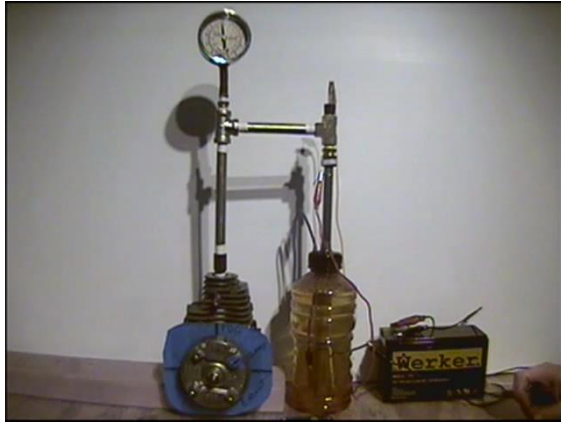
Decreasing supplies of fossil fuels and steadily rising concentrations of atmospheric carbon dioxide concentrations and levels of atmospheric pollutants are some of major challenges to the modern society. The scientific community is addressing these problems by an attempt to replace fossil fuels with cleaner and renewable sources of energy. The research conducted so far indicates the biomass-based fuels to be the best option because they do not require changes in the existing technologies in use. Probably the best alcohol that can be an alternative to petroleum is ethanol. Thus a new path has been opened for bio-fuel engines, i.e. engines that can operate with gasoline blended with anhydrous ethanol (E5-20% vol/vol, 100% hydrous ethanol (4.0-4.9% vol/vol of water) or any blend of these fuels (Shin et al., 2012). The goal of this study was to elucidate the hypothetical synergistic combustion effects of various fuel mixtures including gasoline, ethanol,

ISSN: 2231-5381 <http://www.ijettjournal.org> Page 358

2 Stroke-(from 4) HHO Vacuum Theory Animation

HHO Is Ideal Fuel to ICE

Recent study shows that the HHO explosion and implosion will happens in a complete burning process, which will make the atmosphere pressure can to generate power and will cause the COP (efficiency) of the HHO engine higher than 100%, it will make the engine self-running to be reality. (**need verification**)



4 Videos-HHO explosion-implosion effect

HHO Is Ideal Fuel to ICE

Hybrid Powered Motor Vehicle
Team Lead - M Anand
Member- R Dharmaraju
College Name - Bharath University
Contact Number - 9042242373

Video-E-battery---HHO Gas---ICE---motor

Battery-ICE-Motorcycle



Video-E-battery---HHO Gas---ICE---negative pressure---motor

Air Energy-ICE-Motorcycle



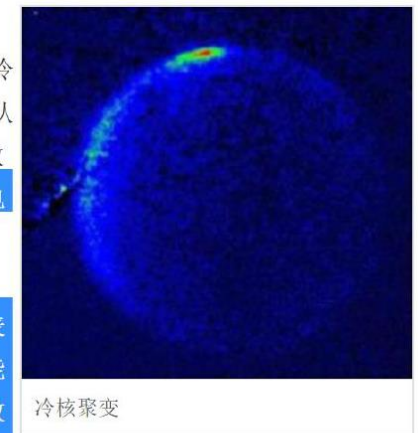
Video-E-battery---COP>1---HHO Gas---ICE---motor

Water Fuel-ICE-Motorcycle



Video-Drinking birds

究。但是，困扰科学家们的問題是如何解释这些异常现象。由于实验用的是重水和钨电极，这很容易给人以错觉，认为这种电化学异常现象就是核现象。因此大部分研究者都在寻求“冷核聚变”的核解释，但在理论上遇到了强大的阻力，进展甚微。另一方面大多数的理论解释认为“冷核聚变”是一种体效应，没有分析电极微观结构的作用。北京航空航天大学的江兴流教授基于实验结果，以电极的尖端效应为突破口，分析了电化学异常现象。江兴流科研组在电解实验中，观察到在电极附近有高度定向的核反应，以及过热、核嬗变、滞后效应（heat after death）。经过不断的探索总结出：气体放电、真空击穿及液体中的放电（电解）现象，有着共同的物理规律，即由于电解过程中电极表面尖端效应产生的聚能过程，在电极表面局部产生气泡和涡旋运动，气泡的产生和坍塌过程将发生动态卡西米尔效应而提取零点能并以热能的方式释放出来；同时涡旋运动与零点能形成挠场相干而提取零点能，一方面释放热能，另一方面形成类星体涡旋结构，在涡旋中心产生高能射线、中子和高能粒子，并伴有高度定向的核反应。可见电极表面尖端处形成远离平衡态的非线性体系，满足一定的条件就会形成自组织的正反馈涡旋，通过挠



冷核聚变

HHO Is Ideal Fuel to ICE

No	Origin	Name	Volume (m3H2/h)	Pressure Bar	Energy Consumption (kWh/m3)H2	efficiency (%) (100%=3.54KW)
1	Norway	Hydro-normal Pressure cell	50-377	1	4.1	86.3
2	Norway	Hydro-normal Pressure cell	10-65	12	4.8	73.8
3	Switzerland	Bamag normal Pressure cell	3-330	1	4.2	84.3
4	Switzerland	Lurgi high Pressure cell	110-760	32	4.3	82.3
5	Belgium	Hydro-normal Pressure cell	15-200	25-Oct	4.2	84.3
6	Switzerland	AccaGenSA cell	1-100	30-Jun	4.8-4.4	73.8-80.4
7	Denmark	Erre Due cell	32-64	4	5.1-5.0	69.4-70.8
8	Russia	Uralkhimmash cell	4-250	10	5.3-5	66.8-70.8
9	USA	HM Hydrogen cell	2.8-11.2	10	~5.5	~64.4

Get HHO from water electrolysis
Electricity Power Extension



H2+O2

The efficiency of getting HHO by electrolysis is same as the conventional way

No	Origin	Name	Power (KW)	Gas production (L/H)	Energy Consumption (kWh/m3)2H2+O2	2H2+O2 efficiency (100%=2.36KW)
1	China	OKQYHJ100	0.3	100	3	78.7
2	China	OKQYHJ200	0.6	200	3	78.7
3	China	OKQYHJ300	0.9	300	3	78.7
4	China	OKQYHJ400	1.2	400	3	78.7
5	China	OKQYHJ600	1.8	600	3	78.7
6	China	OKQYHJ800	2.4	800	3	78.7
7	China	OKQYHJ1000	3	1000	3	78.7
8	China	OKQYHJ2000	6	2000	3	78.7
9	China	OKQYHJ3000	9	3000	3	78.7
10	China	OKQYHJ5500	16.5	5500	3	78.7
11	China	OKQYHJ7500	22.5	7500	3	78.7
12	China	OKQYHJ10000	30	10000	3	78.7
13	China	OKQYHJ20000	60	20000	3	78.7

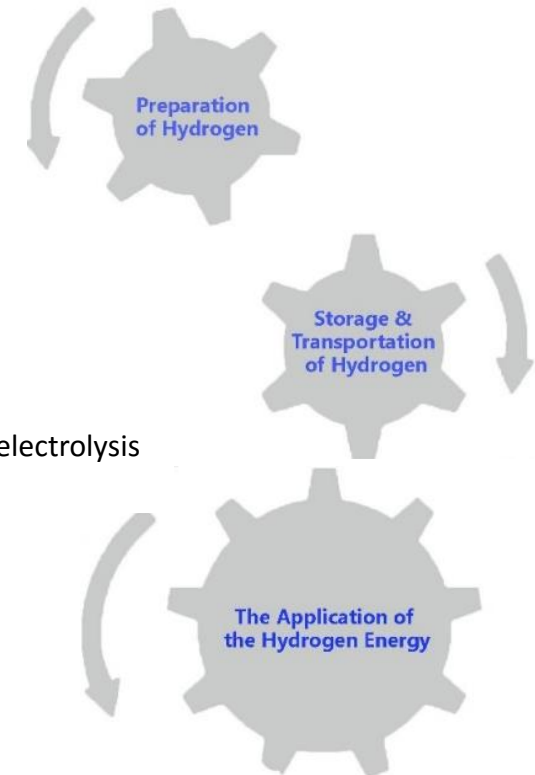
No	Origin	Name	Power (KW)	Gas production (L/H)	Energy Consumption (kWh/m3)2H2+O2	2H2+O2 efficiency (100%=2.36KW)
1	Taiwan	EP-230	1.65	504	3.3	72
2	Taiwan	EP-230C	1.65	504	3.3	72
3	Taiwan	EP-350	2.3	840	2.7	86
4	Taiwan	EP-350B+	2.3	840	2.7	86
5	Taiwan	EP-500	11.5	2688	4.2	55



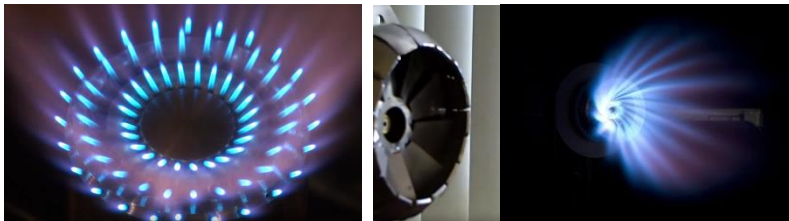
HHO



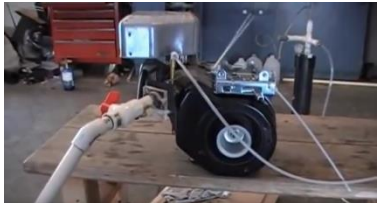
HHO



HHO Is Ideal Fuel to ICE



Burner



Engine



Electric ICE Generator



Vehicle



Motorcycle



Fuel Cell Electric Generator

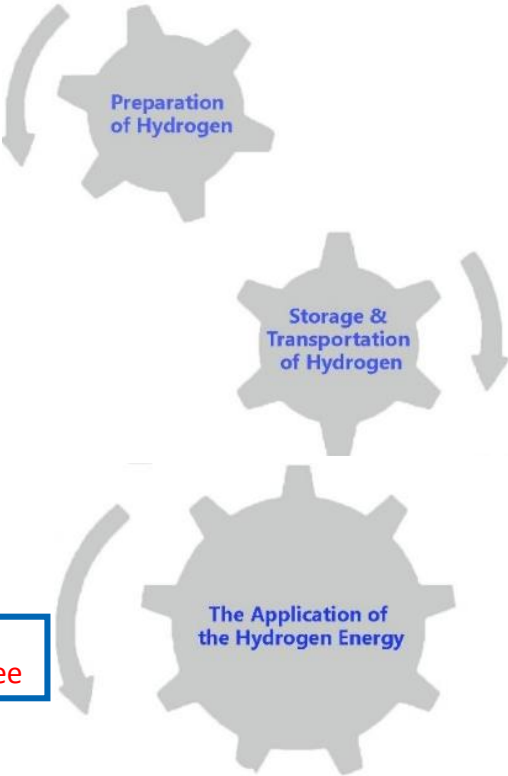


Video-HHO Car Engine Idling(1.3L)



Video-HHO motorbike (35CC)

Get HHO from water electrolysis
Electricity Power Extension



Preparation
of Hydrogen

Storage &
Transportation
of Hydrogen

The Application of
the Hydrogen Energy

MHHO Is Ideal Fuel to ICE

1. HHO - ideal Heater .

- CANNOT be used as a **single fuel** - only **BLENDED**, up to 50%, with gasoline, diesel, methane, LPG, for 20-60% mileage improving.
 - CANNOT be compressed over 2 bar - explodes.
 - CANNOT be stored in bottles.
 - Significant Hydrogen **LEAKAGE** observed through regular sealants.
 - CANNOT be liquefied - oxygen crystallizes.
- Extremely **FLAMMABLE** - CAN be ignited by a small spark.
- VERY LOW own FLAME temperature : **140 deg.C** only.
- ULTRAHIGH (>5,500 deg.C), unique **plasma-flame** performance while heating UHTC ceramics.
- **Ionized** (electric conductive) gas flame, with **PLASMA** properties.
- EASY to produce out of **water**.

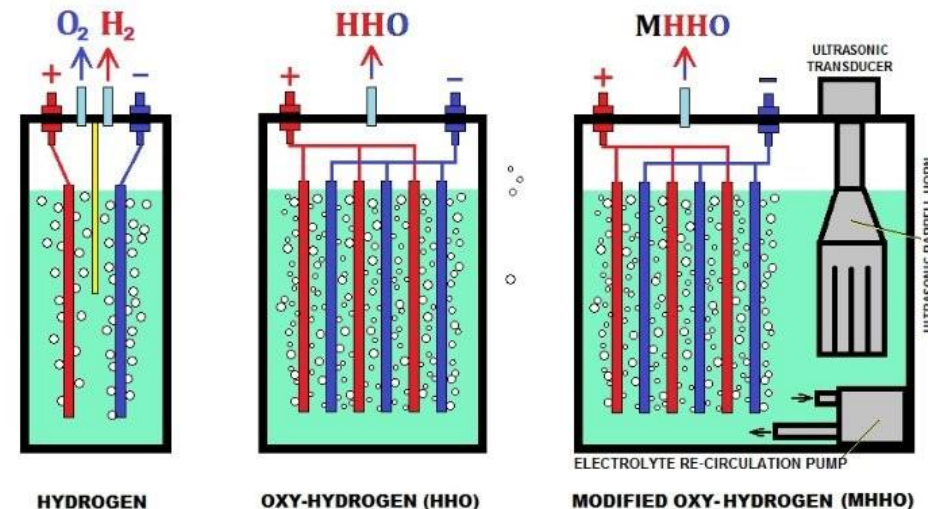
2. MHHO - ideal Fuel.

- CAN be used as a CLEAN and CHEAP **SINGLE FUEL** for ICEs, Gas-Turbines, Plane and Rocket Engines and **BLENDED** (20-30%) with **Pure Hydrogen**, or hydrocarbon gases.
 - CAN be COMPRESSED over 200 bar and safely STORED in bottles for years
 - NO HYDROGEN or else LEAKAGE or Embrittlement observed at all.
 - CAN be LIQUEFIED at -178deg.C/1bar (close to the LNG temp) and stored in LNG bottles.
 - NON-FLAMMABLE - cannot be ignited by a spark at normal conditions - only under pressure and heat.
- VERY LOW own FLAME temperature: **-140 deg.C** only.
- **ULTRAHIGH** thermal performance (5,500+ deg/C), while heating UHTC ceramics.
- **Ionized** (electric conductive) gas flame, with **PLASMA** properties.
- EASY to produce out of water (**NOT so easy** as HHO).

3. PURE HYDROGEN - Universal Fuel, Chemical compound, Lift-Power, etc.

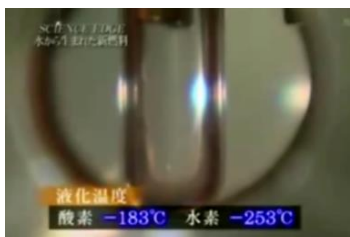
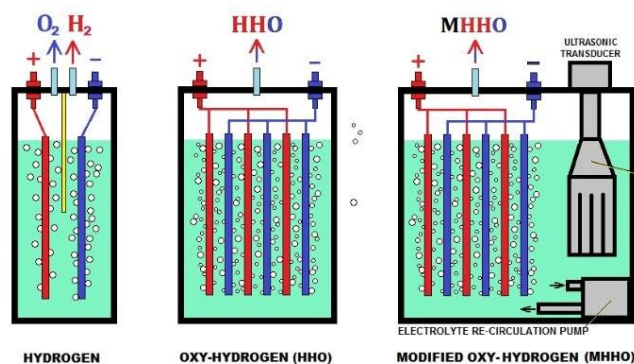
- CAN be used as a CHEAP and CLEAN **SINGLE FUEL** for ICEs, HFCs, Gas-Turbines, Rocket Engines.
- CAN be **COMPRESSED** over 200 bar and STORED in **SPECIALLY ISOLATED** bottles for years.
- **SIGNIFICANT LEAKAGE** and **Embrittlement** properties observed !
- CAN be **LIQUEFIED** at -253deg.C/1bar - very low temp, very **hard** and **costly** to be achieved.
- **EXTREMELY FLAMMABLE**.
- **HIGH** own FLAME temperature : **2,100 deg.C**.
- **Same 2,100 deg.C** thermal performance while heating **UHTC** ceramics.
- **Non-Ionized** (non-electric conductive) gas flame, with NO PLASMA properties.
- Can be used as a fuel in **HFCs** for e-cars onboard power generation.
- Can be used as a universal **LIFTING POWER** (being the lightest element) for Airships and Geostat Sat Platforms
- Can be used as a **feedstock** for hydrocarbon fuels, polymers, etc. production (FT Cycle).
- **EASY** to produce out of water.

HHO VS H₂ VS MHHO



MHHO Is Ideal Fuel to ICE

HHO VS H₂ VS MHHO



	Liquefied temperature	Energy consuming
H2	-253℃	40 MJ/kg (11.1KWh)
N2	-196℃	877KJ/Kg (0.24KWh)
O2	-183℃	3.6-4.5MJ/Kg (1KWh)
HHO	-178℃	0.5~1KWh*
LNG	-162℃	510KJ/Kg (0.14KWh)

COMPANY NAME:
JAPAN TECHNO CO., LTD.
Address:
2-14-10 Kugahara ,Ota-Ku,Tokyo,146-0085,Japan
tel.81-3-3754-1661
fax.81-3-3754-7175
Contact Person
President Ryushin Omasa



TOKYO METROPOLITAN UNIVERSITY



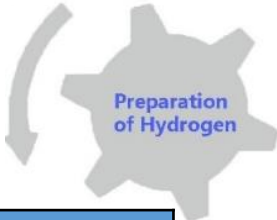
国立大学法人 東京海洋大学
Tokyo University of Marine Science and Technology



Video MHHO introduce

MHHO Is Ideal Fuel to ICE

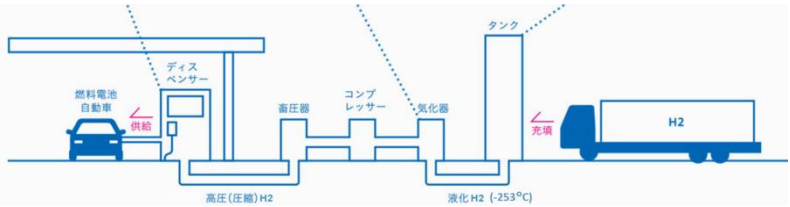
Get MHHO from water electrolysis
Electricity Power Extension



H₂



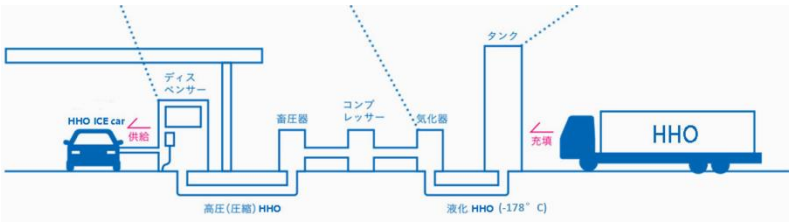
3~5 min



MHHO



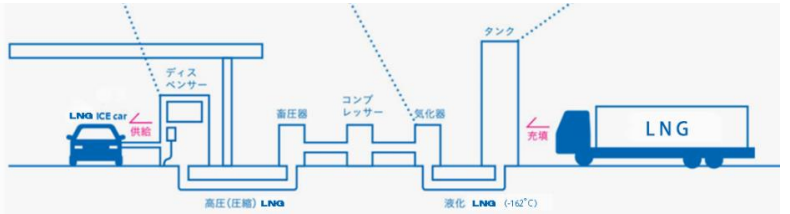
3~5 min



LNG



3~5 min



	Hydrogen Liquid	Hydrogen Gas		MHHO Liquid
	L-H ₂	35MPa	70MPa	L-MHHO
Hydrogen Density	70.8kg/m ³	15kg/m ³	30kg/m ³	167kg/m ³ (converted to Hydrogen content)
				1503kg/m ³ (Liquid MHHO Density)

Liquid MHHO Storage
Use LNG infrastructure



A cubic meter of MHHO contains 2.4 times more hydrogen by mass (167 kg), than a cubic meter of liquid hydrogen (70kg).

MHHO ICE Car

Low Cost, Mature, Durable, Emission free

the HHO liquefied temperature is **-178°C**, it is much higher than the hydrogen and similar to the LNG. As we known, the entire production and storage of LNG is very mature and popular already. We can fully use the existing mature infrastructure. **Humans already have the perfect fuel and the perfect storage. Both could be applied immediately in the existing gas-converted engines.**

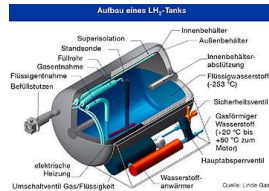


MHHO Is Ideal Fuel to ICE

Hydrogen Energy Industry Old Chain

(Expensive, Energy Consumption, No Mature, No Durable)

- Crack The Liquid Ammonia, Methanol
- Conversion From Natural Gas & Petrol
- Make From The Coal
- Get Hydrogen From Water Electrolysis

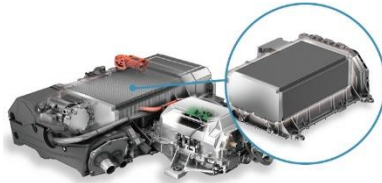


Storage & Transportation of Hydrogen

- High Pressure Storage
- Liquid Hydrogen Storage
- Solid Metal Storage
- Organic Liquid Storage

The Application of the Hydrogen Energy

- Traditional Industrial Hydrogen Consumption
- Hydrogen Fuel Cell (Car, Satellite, Cogeneration)

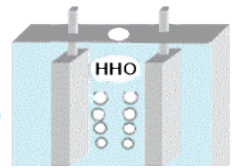


Hydrogen Energy Industry New Chain

(Safe, Low Cost, Mature, Durable, Emission Free)

Get MHHO from water electrolysis
Electricity Power Extension

Preparation of Hydrogen

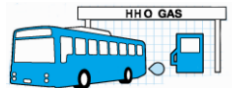


Storage & Transportation of Hydrogen



Liquid MHHO Storage
Use LNG infrastructure

The Application of the Hydrogen Energy



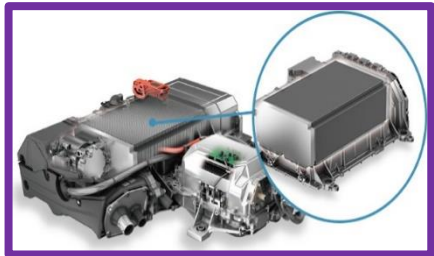
MHHO Engine

Low Cost, Mature, Durable, Emission free

Finding the right hydrogen fuel for the ICE is finding the right hydrogen industry chain for the vehicle.

(找对适合内燃机的氢燃料才能正确打通汽车用氢产业)

H2-ICE Ver2.0 Sole HHO-ICE



Gasoline ICE

H2 ICE

HHO ICE

HFC

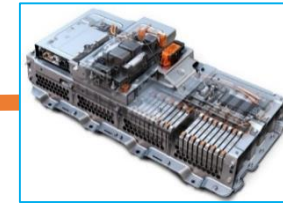
Cox (g/MJ)	89.0	0	0	0
NOx (g/MJ)	30.6	28.8	0	0
Calorific Value (MJ/Kg)	44.0	141.9	352*	141.9 (H ₂)
System Efficiency (%)	20~40	40~47	50~80*	50~80*
Engine Life (10K Km)	30	40	50	50
Size & Weight	Big & Heavy	Big & Heavy	Small & Light	Big & Heavy
Air Quality Requirement	Y	Y	N	Much
Work In Closed Room?	N	N	Y	Conditions
Perfect in 2-strokes?	★★★	★★★★★	★★★★★	
Cost (10K RMB)	2<	>2	2*	20-30(2018) 15-20(2025) 10-15(2030)

★★★★★



HHO

Emission Free ICE Car

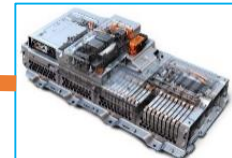


Emission Free New EV Car



HHO

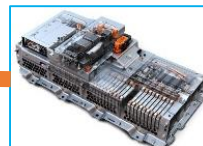
Emission Free Hybrid Car





Methanol

Gasoline



Low Emission Hybrid Car



Methanol

Gasoline



HHO



LNG

Low Emission Four-Flex Car



Methanol

Gasoline



HHO



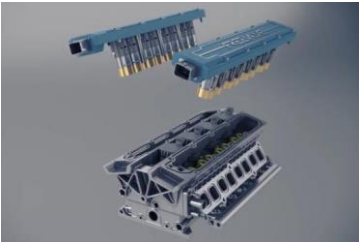
LNG

All-In Car



H2-ICE Ver2.0 Four-Flex Car

If we can design a new engine with the Free valve technology as well as the direct injection. We can make the engine run both in 2/4-strokes mode, in that time, we have chance to just use one **Hydrogen Ready - hybrid Smart Engine** to make a car to meet different market.



This single Hybrid Engine can work as below:

.....Pure HHO Engine Vehicle(**2-Stroke, High Efficient Zero Emission**)

...HHO- Methanol Hybrid Engine Vehicle(**Ultra Low Emission**) <5L/100KM

...HHO- Gasoline Hybrid Engine Vehicle(**Ultra Low Emission**) <5L/100KM

H2- Methanol Hybrid Engine Vehicle(**Ultra Low Emission**) <5L/100KM

...H2- Gasoline Hybrid Engine Vehicle(**Ultra Low Emission**)<5L/100KM

.....Pure H2 Engine (**Vehicle(Green Emission)**)

...Pure Methanol Engine Vehicle(Low Emission)

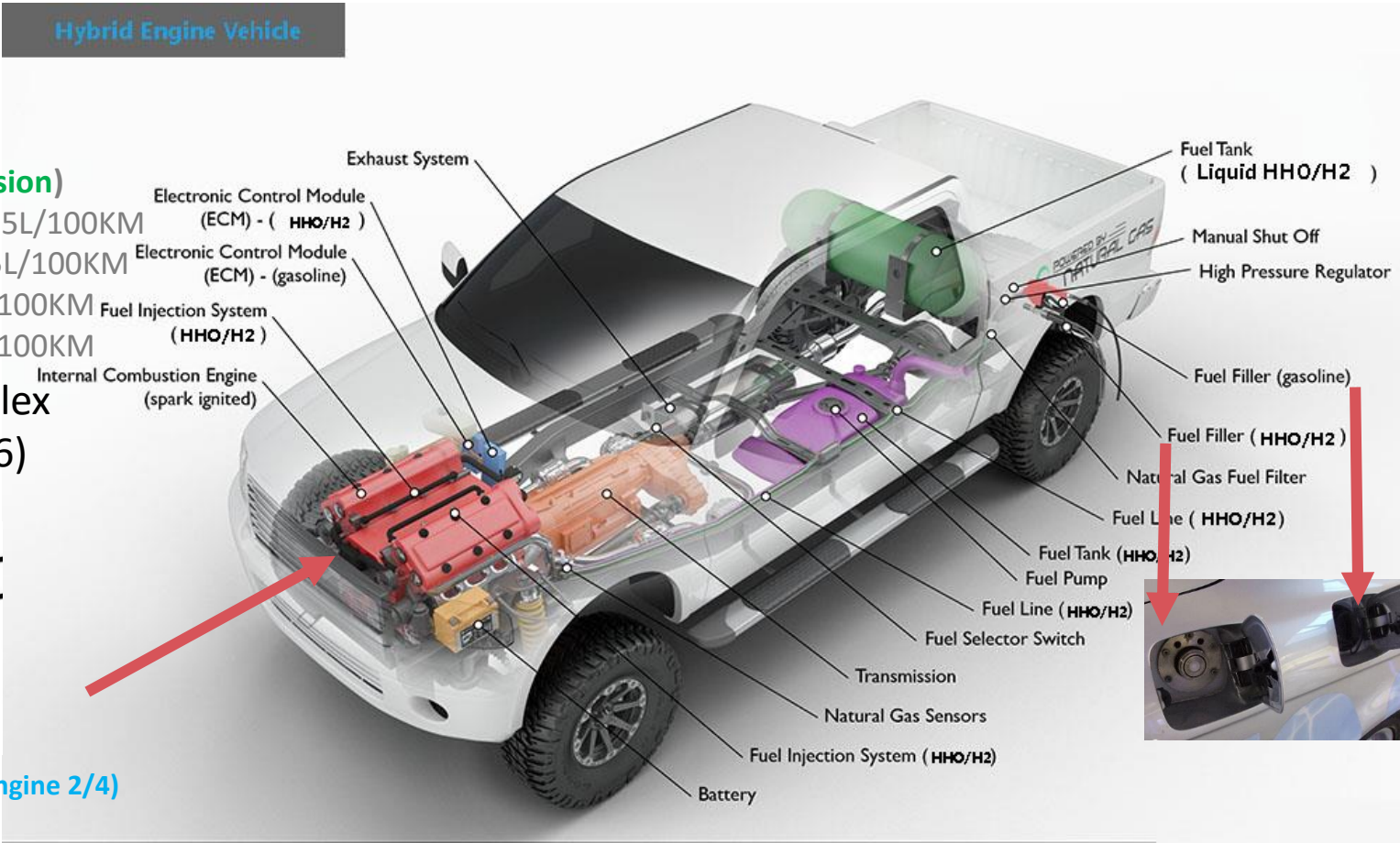
...Pure Gasoline Engine Vehicle(Normal Emission)

.....



H-H Engine
(Hydrogen Ready Hybrid Smart Engine 2/4)

Ford Tri-Flex
V10 (2006)



H2-ICE Ver2.0 Four-Flex

In this scenario, customer select the vehicle that will switch from simple engine displacement options to hybrid engine content options. The customer decides which power distribution he likes, but regardless of the route, the green and practical principles are taken into consideration.



Before

	Zetec	ST-Line	ST-Line X	Titanium	Titanium X
Engine	1.6L	1.6T	1.8L	1.8T	2.0L
			●		

Later

Hydrogen Ready Hybrid Engine (A+B)	A Function	Hydrogen (High Pressure)	Hydrogen Liquid	HHO (High Pressure)	HHO (Liquid)
			●		●
	B Function	Gasoline	E85	M85	M100
		●	●	●	●

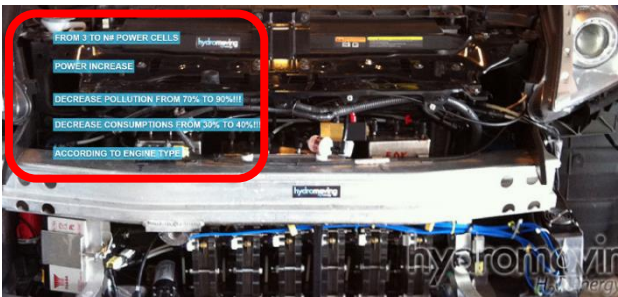


High Pressure VS Liquid tank



HHO 技术商业化的四个技术层次

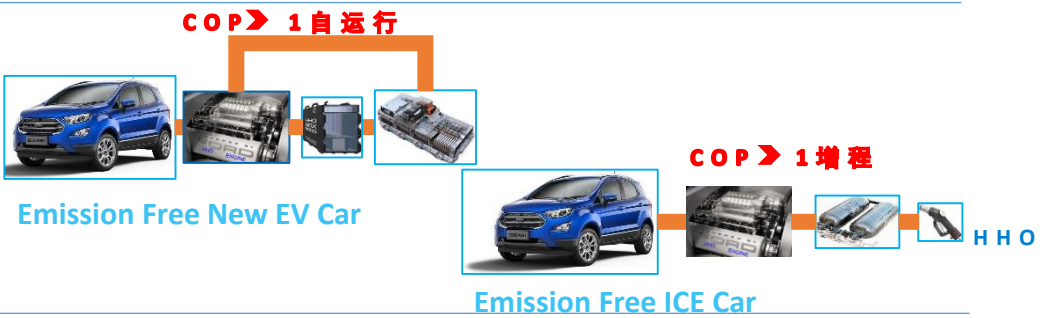
1, HHO Kit, 节油30-40%, 减排70-90%, 某宝有卖, 但未普及开, 在“十二五”期第一批全国重点推广公路水路交通运输节能产品（技术）目录里面, OEM厂可以介入这一块, 后装市场或者前装市场都可以。非汽车界可以在切割行业, 焊接行业, 节能减排行业, 抗灾救险, 医疗美容行业大展拳脚, 这些行业现在正是这个技术主要推广使用范围, 契合国家节能减排, 绿色经济发展方向。



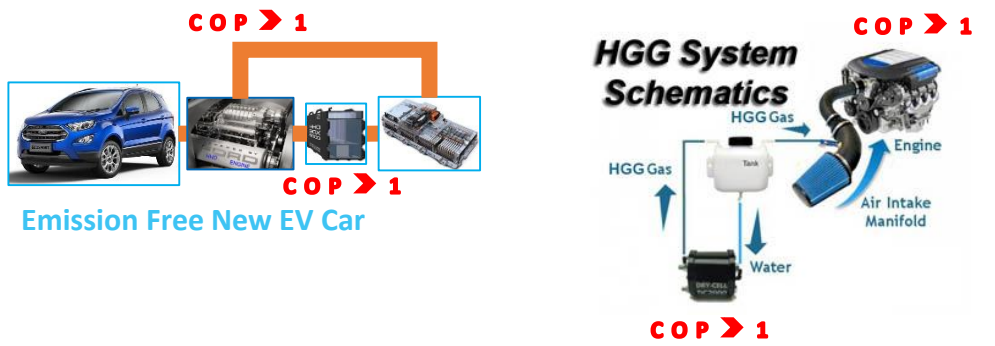
2, HHO Engine, 基于HHO液化确实如天然气一般方便, 高效, 安全, 同时纯HHO发动机效率达到50-80%开发成功。低成本零排放内燃机车PK掉燃料电池车PK掉化学电池电动车情形出现, 同时运载火箭大尺度瘦身, 发射成本极大降低成为可能。这个是近期非常有可能达到的状况。独角兽爆发之地。



3, HHO Engine, 基于HHO发动机内爆负压现象出现, 使提取大气压势能成功, HHO发动机效率80-100%, 假水燃料发动机（假永动机）出现, 汽车只加注水运行, 燃料加注站可以脱网依靠气压势能生产并液化HHO。自由能开始真正出现在大众生活中。能源成本将极大降低。



4, HHO Engine效率超100%, 同时HHO电解制成过程成功提取零点能, 使效率也超100%, 假水燃料发动机（假永动机）大功率作业, 脱网独立零排放汽车大功率发电让汽车成为主要能源站, 承担分布式清洁能源重要一环。人类彻底摆脱石化经济束缚, 经济发展成本与环境友好性得到质的飞跃。



HHO 实验路径清单

A1: 通过高速摄像机, 观察极板间HHO的生成规律。

A2: 测量HHO的产气率与极板间隔, 面积, 与电脉冲参数的关系, 实验不同脉冲电环境下的产气效率 (纯度)。



B1: HHO气体摩尔称重 (测量HHO液化提纯后再汽化状态)

B2: HHO气体标准大气压下的液化点 (如果实验条件许可, 测试加压液化)

B3: 测试HHO气体拉曼光谱吸收 (对比氢气与氧气2: 1成分的光谱吸收)

B4: 测试HHO2: 1摩尔比下面的压燃 (压爆) 参数, 与安全压力储存参数

B5: 测试HHO火焰传播速度, 燃烧过程压力变化与温度变化情况



C1: 纯HHO供发动机运行的调校实验 (两冲程/四冲程)

C2: 基于C1成功的情况下, 跑全工况, 记录相关转速, 输出功率, 扭矩, 燃料消耗量, 实际功率与排放温度和湿度的情况。

C3: 研究内爆现象对发动机功率提升或者获取环境能路径探索。

C4: 根据以上假设实验成功收集信息充分的情况下, 推导出HHO内燃机关键参数。指导下一步纯HHO发动机研发方向。

Go Further

RESEARCH

FRAME INSIGHTS

CONCEPTS / PROTOTYPES

SOLUTIONS

Compare with hydrogen, there are indications that HHO, as a composite form of hydrogen energy, has obvious cost advantages and a mature industrial foundation at the manufacturing, transport and storage, as well as at the automotive energy consumption. It is very likely that HHO will become a flashpoint for new hydrogen energy in the future.

It is recommended that a small project should be set up to build a HHO-ICE model to verify the entire technology.

